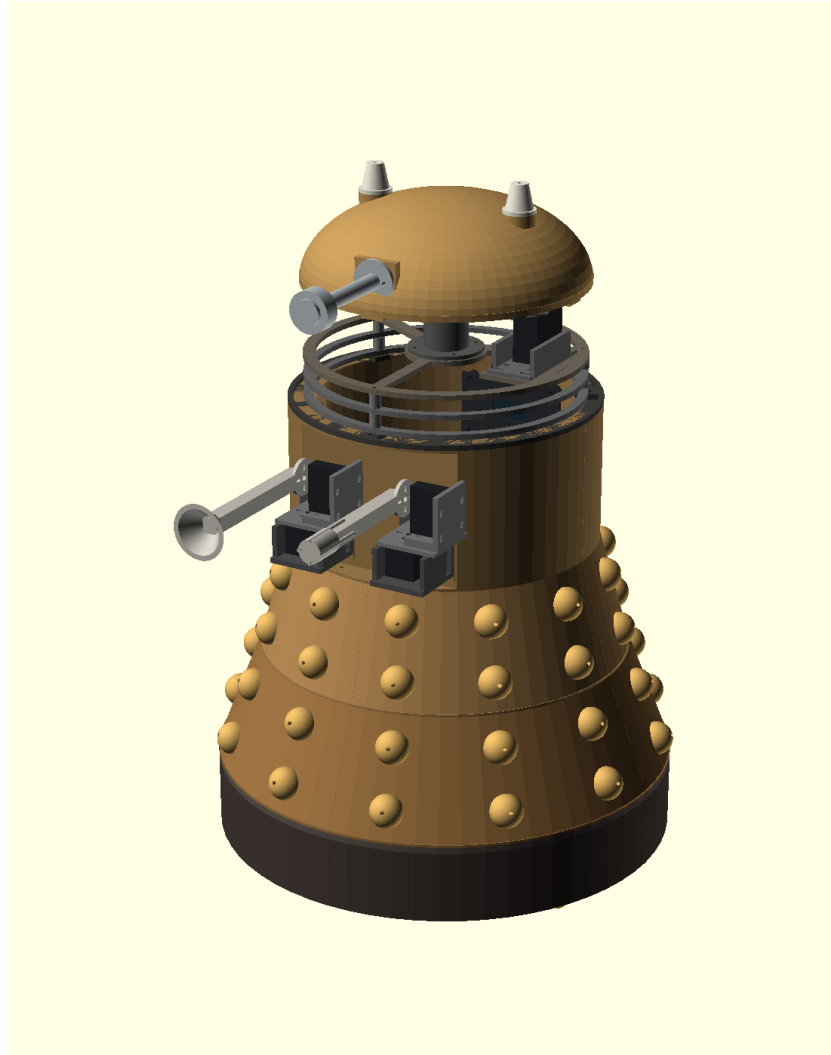


Dalek robot design and assembly

A modular printed robot with four TT drive motors, a rear ESP32 display, local Wi-Fi controls, two arms that trace small circles, and a head that can rotate continuously.

The package includes printable STL files, editable OpenSCAD source, circuit diagrams, mounts and purchased hardware lists, firmware, 24 original MP3 clips, and a staged assembly guide.

This is a digitally checked design for a first prototype. Physical fit, loaded traction, supply temperature and stopping behavior require the tests in this manual. Do not skip those gates.



CAD assembly view. Purchased parts and internal features may be simplified in the preview.

Using this package

Read the assembly sequence first, then use the exact part-specific mechanical instructions. Read the firmware procedure before wiring or flashing. The electrical research chapter explains the power and pin choices. Full-size circuit sheets and complete bills of materials follow the text.

The source package is in JeremyProffittOrg/robots on main under dalek/. Each drawing and source filename in this PDF is relative to that directory. Original files and firmware:
<https://github.com/JeremyProffittOrg/robots/tree/main/dalek> .

Use the PDF bookmarks to jump between chapters. The final appendix lists parts and wires. The supplied digital verification record reports what was actually checked and what remains unverified.

Assembly and commissioning sequence

Build and test one subsystem at a time. The mechanical chapter gives the exact part filenames, fasteners, bearing fits and assembly order. The electrical chapter and circuit sheets give the exact wire connections. This chapter places those operations in a safe order and defines when each stage is complete. A check is not complete until it has been performed on the physical robot.

1. Inventory before printing

Read the whole manual and open the CAD preview. Inventory `bom/electronics.csv`, `bom/hardware.csv`, and `bom/printed-parts.csv`. The quantities in the printed-parts list are quantities to make, not the number of files. Some decorative and structural pieces are used repeatedly. Keep screws in labelled bags by thread and length.

Check the actual board: it must be the original TTGO T-Display ESP32 with the 1.14-inch screen. Check the labels on all five servos, four motors, two driver boards and battery. Do not fit an S3 board or a same-size-looking servo without checking the mechanical drawings and pin assignments. The selected wheel and head servo were listed out of stock during research; obtain those exact parts before declaring the build ready to assemble.

Tools: a digital calliper, metric hex keys, small Phillips drivers, flush cutters, a deburring tool, soldering iron, solder, wire stripper, crimp tool for the selected connectors, heat-shrink tube, digital multimeter, a current-limited bench supply, and a scale. A contact temperature probe and an oscilloscope improve the power and noise tests. Use eye protection when cutting wire or drilling. A threaded fastener, bearing, motor, battery or connector is purchased hardware, not a printed substitute.

Done when every required purchased item is present or its absence is recorded, the board variant is confirmed, and the tools can measure voltage and current. This is an inventory gate. It does not require assembling or energising anything.

2. Prove fit with small prints

Print the fit coupon and critical holders first. Use the intended structural material and slicer profile. Check screw-clearance holes, nut pockets, motor fit, servo fit, bearing seat and the T-Display opening using the real hardware. Clean a first-layer edge before concluding that a dimension is wrong. Do not force a bearing or crush a board into a tight print.

If a fit needs adjustment, change the relevant clearance parameter in the OpenSCAD source, export that part again, and test it again. Do not scale the whole model in the slicer. Record the material, nozzle, layer height and clearance that passed. A printer-specific compensation may be needed, but it must preserve the joint centres and part lengths.

Done when fasteners pass without splitting the part, nuts retain their flats, the motor cannot shift in its holder, servo bodies seat without force, and the screen is visible with the USB connector accessible. The rear holder must retain the PCB by its edges without pressing on the glass.

3. Print and label the remaining parts

Follow the orientation and quantity fields in `printed-parts.csv`. Start with the chassis, brackets and joining parts. Then print the skirt, shoulders, neck and dome. Use the settings in the mechanical chapter. Keep the supplied model at 100% scale. The delivered orientation is part of the bed-fit check; inspect each part's contact face before slicing.

Use the H2D profile for the nozzle actually installed. A single material can make the whole robot, with paint for the bronze and black surfaces. Colour changes are cosmetic and should not affect a joint. Add a brim where the print notes call for one. Remove supports fully from nut traps, screw holes, grooves and bearing seats. Keep paint out of moving fits.

Weigh each group after support removal. Record total printed mass and the eventual fully assembled mass. Do not use slicer infill percentage as a direct mass estimate for thin shells: most of a thin shell may already be walls. The validation report's solid mesh volume is a useful conservative bound for fully filled material, not a slicer timing estimate.

Done when every manifest quantity is made, identified, free from cracks and checked against its mating piece. Do not close the cosmetic shell yet.

4. Assemble the chassis without power

Follow the chassis and motor-holder steps in `mechanical1.md`. Fasten printed joins with the listed bolts, washers and nuts. Tighten until the joint is secure; further tightening can crush printed plastic. Do not put liquid thread-locker on printed plastic. Use the locking hardware in the bill of materials.

Seat all four motors in the specified direction. Give their thin motor leads strain relief. Press each wheel onto its TT output shaft while supporting the gearbox body. The wheel's hub fit provides retention; do not drill the motor shaft. Check that tires cannot rub the bumper, a bolt, or a motor cable at any rotation. Check that the assembled chassis does not rock on a flat surface.

Install the empty battery cradle and its straps. The battery remains disconnected. Keep the pack low and centred. It must not rest on screw points or sharp layer edges. Mount the electronics deck and insulating standoffs. Keep the deck removable until commissioning is complete.

Done when all wheels rotate by hand without body contact, motors cannot slide, the chassis rests evenly, and the battery can be removed without cutting wires. The mechanical chapter defines the exact floor and wheel clearances.

5. Build the power harness on the bench

Make the harness from the circuit sheets and `electronics/wiring.csv`, using the listed wire gauges and connector ratings. Put the main fuse close to the battery connector. Put the physical actuator switch in the feed to both movement converters. Keep the logic branch separately powered from the main switch. Terminate returns at the common ground point. Do not use a solderless breadboard or loose Dupont contacts for motor, servo or battery current.

Label every connector at both ends. Use a different keyed connector, or a clearly different connector size, for battery voltage and 5 V. Insulate exposed terminals and add strain relief. Confirm electrolytic capacitor polarity. Fit the motor noise capacitors close to the motor terminals. Never place a diode directly across a bidirectional motor as though it were a relay coil.

With the battery disconnected, check continuity of every wire against the net list. Check for shorts between each supply and ground. Inspect the physical switch and fuse ratings; an AC-only current rating is not a sufficient DC switch rating. Check the ADC resistor values with the meter before connecting an ESP32 input.

Done when every net has been checked, no supply is shorted, the battery connector polarity is correct, and the actuator switch interrupts both movement branches. Keep all downstream boards disconnected for the next stage.

6. Check the three supplies before connecting boards

Energise the harness from a current-limited bench supply at a voltage in the battery's operating range. Check each converter's unloaded output. Each 5 V branch must be within its specified tolerance and suitable for the attached devices. Then test each branch with an appropriate known load. Do not connect the battery voltage to a 5 V rail to make a failed converter test pass.

Confirm the actuator switch makes the motor and servo outputs fall to zero after stored capacitor charge has decayed. Confirm the logic output remains available. Check the external pack divider voltage against the calculation in the electrical chapter. At the maximum charge voltage it must remain below the ESP32 input limit. A reversed divider is a board-damaging error.

Done when the motor, servo and logic rails pass unloaded and loaded checks, the pack divider passes, and the physical actuator switch has the intended effect. A warm regulator alone does not prove a safe load; measure its temperature and voltage under load.

7. Flash and start the controller on USB only

Disconnect the battery and external controller 5 V supply. Leave all actuator supplies off. Connect the T-Display to the computer using USB. Do not back-feed the USB supply from an external 5 V source. Use the complete flashing procedure in `firmware.md`, including both the program and the LittleFS image. A program upload alone does not install the audio files.

After boot, check that the rear display is readable and shows network and safety status. Join the robot's access point using the credentials shown by the controller. Open the displayed local address. Check that the interface loads with the phone's Internet access disabled. It must display the sound list and controls without fetching a web font or script from a public website.

Store home Wi-Fi settings only through the local setup form. Use a 2.4 GHz network supported by the ESP32. Check the displayed station address after joining. If the home network is unavailable, follow the fallback procedure. Do not put Wi-Fi passwords in source files, a screenshot, this manual, or Git.

Done when the firmware and filesystem load, the TFT is readable, direct AP control works, station setup works on the intended network, and startup does not arm movement. Hardware-dependent status may correctly block arming until the next stages are complete.

8. Verify logic and hardware disable states

With USB removed, reconnect the external logic supply and the low-voltage control harness. Keep movement power off. Confirm the ground is common before connecting signal wires. Check the PCA9685 address and the buffer connections. Verify the motor sleep line is low and the servo output-enable line is in its disabled state at reset.

The servo buffers operate from the switched servo supply. Each PWM path, buffer output and ground must match the circuit sheet. The servo power cable must never connect to the ESP32's 3.3 V pin. Check the ADC display against a meter and apply the documented calibration factor if needed.

Done when reset leaves actuators disabled, the controller detects the actuator switch state, the PCA9685 responds, and battery voltage agrees with the meter to the stated tolerance. A blocked arm command is the correct outcome while power or sensor checks fail.

9. Test drive direction with wheels lifted

Support the chassis so all tires are clear of the bench. Keep hands away from the wheels. Connect the movement supplies and explicitly arm through the interface. Apply a brief, low command to both sides. Each wheel must turn in the direction that would move the robot forward. The right-side motors are physically mirrored, so their lead polarity must be checked, not copied by wire colour.

If one motor on a side turns backward, disarm, isolate power and reverse only that motor's two output leads. Do not connect two driver outputs together. Once directions are correct, test a slow forward command, a stop, a slow reverse command and a stop. Listen for rubbing and inspect driver temperature. Do not hold a stopped wheel against power to measure stall behavior.

Done when all four directions are correct, release stops the command, reversing passes through zero, and no tire or wire rubs. Leave final turning and loaded traction tests for the floor stage.

10. Centre servos before attaching the arms

Keep the arm horns and head drive connection detached. Command the positional channels to their documented centre pulse. Check that each servo stops and does not buzz against a hard limit. Fit the supplied horns at the neutral positions shown in the mechanical chapter. Retain each horn with its own centre screw. Do not print a replacement servo spline.

Assemble each gimbal using its exact brackets and retaining hardware. Route the servo lead with a free loop for the full small-angle movement. Do not tie that loop across a hinge. Fit the lightweight arm shaft and decorative end. Check by hand that each axis can move through its intended range before applying power.

Start the circular motion at a small radius and low frequency. Check one arm at a time. Correct a reversed axis using the calibration setting, not a cable swap on a three-wire servo. Increase only to the supplied bounded operating range. The tip should trace a near-circle without the arm touching the shoulder or pulling a wire.

Done when both arms can complete repeated circles quietly, every fastener remains secure, no servo stalls, and the supply stays within tolerance. Continuous buzzing, a hot servo or a distorted orbit calls for a mechanical or calibration correction before proceeding.

11. Assemble and test the head

Follow the bearing, shaft, belt and retention sequence in `mechanical.md`. Check all bearing seats before fitting the dome. Turn the assembly through several full revolutions by hand. Check the eye-stalk sweep against the arms and the surrounding area. Keep the head drive lightly tensioned as specified; excessive belt tension loads both the servo and bearing.

With the drive disconnected, adjust the continuous servo's neutral setting until it stops. Connect the drive and test low speed in each direction. The rotating structure carries no wires. Secure the fixed servo lead below the rotating interface. Fit the final dome fasteners only after checking that no screw touches the moving ring.

Done when the head turns repeatedly through full revolutions in both directions, neutral stops creep, and the dome remains retained when gently lifted by its intended service points. Do not use the eye stalk as a lifting handle. The controller sets head speed, not a measured head angle.

12. Install and check audio

Fix the speaker to its body mount with the specified retainer. Keep both speaker terminals isolated from chassis ground. Wire the MAX98357A exactly as shown; its bridge speaker output has no grounded speaker lead. Confirm the gain resistor before applying power.

Play `hello.mp3` at low gain. Then test every sound button. Check for distortion, controller resets and motion pauses while audio plays. Keep gain within the speaker's 1 W limit and the logic supply budget. Warning phrases in the collection are available as sound effects; their filenames do not prove automatic event playback.

Done when all 24 files play on demand, speech is intelligible, sound does not reset the ESP32, and amplifier/speaker temperature stays acceptable during repeated use. Keep the speaker and amplifier wiring fixed in the body.

13. Run the stop tests before floor driving

Keep wheels lifted for this complete test. Arm and command slow motion. Release the drive button and confirm the command stops. Repeat and change browser tabs. Repeat and close the page. Repeat and disable the phone's Wi-Fi. Repeat and turn off the access point or station connection used for control. Measure the delay to removal of the drive signal; the firmware lease is 500 ms, and the exact task and I/O delay must be checked on hardware. Tire coasting after removal of power is not the same as a continued drive signal.

Turn the actuator switch off during a slow command. Both drive and servo power must be removed. Turn it on again. Movement must not resume until the operator explicitly arms again. Reset the ESP32 during a slow command and verify the same no-restart behavior. Check that a second browser cannot revive a stale command after a stop.

Test low-battery detection with a bench supply, not by repeatedly exhausting the actual battery. Follow the voltage thresholds in the electrical and firmware chapters. Restore normal voltage and confirm recovery still requires explicit arming. Temporarily disconnect the servo-controller communication while safely supported, and confirm the documented fault response. Reconnect with power isolated.

Done only when every stop case passes. Record the measured time and test setup. Do not move to floor testing with any unexpected restart or uncontrolled motion.

14. Close the shell and run a short floor test

Install the remaining panels in the mechanical sequence. Check the rear screen opening, switch access and service access. Tie fixed cables clear of wheels, arms, neck rotation, fastener tips and ventilation. Fit strain relief to removable-panel cables. Tighten the battery straps and perform a gentle push test to confirm the pack cannot shift.

Weigh the complete robot. Begin on a clear, level hard floor with an operator beside the physical stop switch. Use low speed and broad forward arcs. Do not begin with a stationary pivot, carpet, ramp or threshold. The specified tires can create more steering resistance than these small motors can overcome. Stop immediately if a wheel stops turning while commanded.

Run for one minute, isolate power and inspect joints, cables, tires, motor temperature and converter temperature. Repeat for five minutes, then fifteen minutes if the earlier test passes. Record battery voltage, mass, floor type, speed setting, current and temperatures. A conservative commissioning gate is to stop testing at 50 C measured on a motor, servo or nearby printed mount, or on any sign of softening, odour or repeated electrical shutdown. This is a build-test threshold, not a replacement for manufacturer component ratings.

Done when the fully dressed robot drives, steers in moving arcs, moves both arms, turns the head, plays audio and stops correctly under the same measured load. This is the first physical proof of the design's operating envelope.

15. Charge, maintain and store

Switch off and disconnect the robot from the battery before charging. Use the specified Bioenno charger on the pack's charging connector. Do not charge through the ESP32, an Adafruit single-cell charger, or a converter output. Charge on a stable nonflammable surface with ventilation. Follow the battery manufacturer's limits and instructions.

Before each use, check the straps, wheel fit, arm horns, head retention and exposed wires. Test the physical stop switch and release-to-stop behavior. Stop use if the battery is damaged or swollen, a connector is discoloured, a motor binds, or a printed bracket cracks. Replace a damaged structural part rather than gluing across a loaded fracture.

Do not carry the robot by an arm, the eye, dome decoration or removable panel. Lift from the chassis with battery power isolated. Keep the wheels free of hair and thread. Store the battery disconnected according to its manufacturer's guidance. Keep a copy of the calibrated firmware settings with the robot.

Fault guide

- Screen works but movement is blocked: check actuator switch state, pack ADC calibration, PCA9685 response, explicit arming and command timeout. Read the displayed fault before changing code.
- Controller resets when a motor starts: check rail sag, common ground, motor capacitors, converter load and connector resistance. Do not disable brownout protection.
- One wheel turns backward: isolate power and swap that motor's two bridge leads. Check all four wheels again while lifted.
- Arms buzz or fail to circle: check horn centres, direction calibration, gimbal contact, servo supply and 5 V signal buffers. Reduce amplitude while diagnosing the fit.
- Head creeps at stop: recalibrate the continuous servo neutral. If it binds only at one angle, inspect bearing alignment, fastener intrusion and dome clearance.
- Robot moves straight but will not pivot: use moving arcs and a lower-friction floor. Measure mass and current. Do not raise motor voltage or remove current limiting.
- Audio is missing: confirm both firmware and LittleFS were flashed, inspect the sound list, verify I2S pins, and check speaker wiring and amplifier gain.
- Home network fails: join the local AP and follow the firmware recovery instructions. Reconnection must not restart motion.

Physical test record

Record date, builder, firmware commit, slicer profile, material, loaded mass and floor type. For each stage above, write PASS or FAIL with the measured result. Include supply voltages under simultaneous load, pack-voltage calibration error, stop delay, fifteen-minute temperatures and observed runtime. Until that record exists, the package remains a digitally checked design for a first prototype.

DALEK-541 mechanical design and assembly

This is a complete printable prototype design. It uses four specified Adafruit TT motors and four specified wheels. It is 541 mm (21.3 in) tall. The bumper is 360 mm in diameter. The eye and arms extend beyond the body. The rear panel exposes the original TTGO T-Display screen, its buttons, and a USB-C cable opening. Two independently controlled joints on each arm trace small circles. A separate bearing-supported head turns continuously. No wire enters the rotating head.

The supplied CAD and STLs are real solids, not a visual concept. Physical fit, traction, print strength, and endurance still require the listed bench tests. No robot has been printed or driven during this design run. Do not treat a passed mesh check as a hardware test.

Files and verified limits

- `cad/dalek.scad`: editable parametric source; millimetres. Set `part="assembly"`, `part="section"`, or `part="exploded"` to inspect it.
- `stl/`: 35 distinct meshes. `bom/printed-parts.csv` specifies 151 printed pieces including 32 small PCB spacers and one fit coupon.
- `cad/assembly.png`, `cad/rear.png`, `cad/section.png`, `cad/exploded.png`: rendered CAD views. Purchased parts shown as simple blocks are envelope references.
- `cad/validation.json`: all 35 meshes pass watertight, consistent winding, positive-volume, one connected positive solid, and 300 x 300 x 300 mm checks. Negative internal cavity surfaces in the arms and eye are allowed. They are sealed air cavities, not loose parts.
- The combined solid material bound is 1988.1 g for every BOM quantity, including the fit coupon and spare spacers. This is geometry multiplied by 1.24 g/cm³ PLA or 1.27 g/cm³ PETG. It is not a measured print mass. The installed print mass will depend on the slicer. Supports and waste are excluded from this number.
- The design targets a loaded mass at or below 3.0 kg. The 0.7 kg battery, purchased electronics, motors, wheels, bearings, and metal fasteners must be added to the actual sliced print mass. Weigh the finished machine. The solid bound alone does not prove the loaded target. Do not add ballast.

Official H2D specifications give 325 x 320 x 325 mm for one nozzle and 300 x 320 x 325 mm for dual-nozzle printing. Every exported part is smaller than 300 mm on every axis. Leave room for a brim and do not fill the bed edge with support towers. [Bambu Lab H2D specification sheet](#).

Researched component interfaces

Adafruit 3777 is a 3-6 V TT gearmotor. Its nominal body is 70 x 22 x 18 mm. The manufacturer drawing shows a 22.44 mm body height, 18.60 mm gearbox thickness and 36.60 mm overall span across the two axle ends. The output axle is near one end of the body. It must remain free on both sides. The cradle uses padded body retention and two cable ties; it does not depend on uncertain case screw holes. [Product, dimension drawing](#).

Adafruit 3766 is a press-fit TT wheel with a 63 x 29 mm envelope. Each weighs 38 g. It was listed out of stock when researched. Keep this wheel selection; another diameter changes clearance and speed. There is no encoder. [Adafruit 3766](#).

The four arm servos are Adafruit 2307 TowerPro MG92B. The listed envelope is 36 x 12 x 31 mm including the mounting features. Use the supplied 20-spline horns and centre screws. The printed parts deliberately do not copy the spline. The manufacturer requires 5 V control signals as well as servo power. The electrical design supplies

this. [Adafruit 2307](#).

The head servo is Adafruit 154 FeeTech FS5103R, listed at 37 x 54 x 20 mm with a 25-spline output. It is a speed servo, not an angle servo. It was listed out of stock when researched. The slotted cradle and foam pads allow the supplied horn to be aligned with the belt plane. The belt and bearings carry the design load path; the servo shaft does not support the dome weight. [Adafruit 154](#).

The rear cassette is for the original ESP32 TTGO T-Display, not the T-Display-S3 or an AMOLED variant. LILYGO documents the 1.14 in display, two buttons and USB-C interface. The printable pocket is 52.2x26mm; the design uses a nominal 51.52x25.04mm board envelope. This dimension is a fit assumption until the actual board is measured. If a revision exceeds the pocket, change the pocket and reprint the cassette before printing the rear body. [LILYGO documentation](#), [original board design files](#).

The speaker is Adafruit 1313, the 77.8x77.8x25.49mm³ in 8-ohm model. Its mounting tabs use 60 mm spacing. The bracket provides a 69 mm sound opening and a 90 mm square plate. [Speaker](#), [manufacturer drawing](#).

The battery tray has 116 x 72 mm clear plan space, 15 mm retaining walls, two 20 mm strap passages and a 36 mm rear connector opening. It accepts the selected Bioenno BLF-1206A6Ah pack with padding. Reserve at least 30 mm behind the battery connector and keep the whole harness below the electronics deck. The battery is removable for charging. Follow the electrical document for the pack and charger; do not connect the 12 V pack to the T-Display battery socket.

Coordinate and joint map

The floor is Z0. Positive Y is the rear. Positive X is the right side. All distances below are in millimetres.

Interface | Location | Retention

Wheel centres, nominal | X±130, Y±65, Z31.5 | Wheels press on the specified TT shafts

Motor cradle centres | X±100, Y±40, Z17 | Four M3x65 bolts per cradle

Chassis | Z72-77 | Four flat quarters; 12 splice plates

Battery tray | Z77-92 | Four bolts at X±45, Y±30

Electronics deck | Z172-175 | Four 95 mm posts at X±70, Y±50

Lower skirt | Z77-167; R180 to R145 | Four quarters; M3 seams and flanges

Upper skirt | Z167-257; R145 to R120 | Four quarters; removable rear quarters

Shoulder | Z257-367 | Front and rear halves; full front transition ledges

Arm bracket backs | Y-102; bases Z270 | Four bolts each; centres X±43

Arm yaw axes, nominal | X±43, Y-129, Z305 | Supplied servo horns

Arm pitch axes, nominal | X±43, Y-129, Z335 | Perpendicular supplied servo horns

Display centre | Rear Y120, Z323 | Four M3x25 cassette bolts

Switch centres | X±30, rear wall, Z279 | 12.5 mm panel bores; switch mounting nuts

Neck adapter | Z367-371 | Three shoulder bolts at R110

Neck rings | Z371-375 and 386-390 | Six 11 mm spacers; three through bolts

Neck deck | Z401-405 | Three M3x45 neck-stack bolts

Bearing tower | Z405-431 | Four M3x12 deck bolts

Head belt centres | X0 and X72.5; Z449.5 | 3 mm round belt; slotted servo adjustment

Head plate | Z457-460 | Four M3x16 bolts at R17

Dome | Z460-525 | Four M3x10 bolts at R101

Lamp bases | Z519 | Flat pads on dome; two M3x16 bolts

Maximum height | Z541 | Top of decorative lamps

The two axle centres are 130 mm apart fore-and-aft. The approximate track is 260 mm. Actual press-fit depth affects the final track. Keep both sides equal. At the nominal position, the inner wheel faces are $X \pm 115.5$. Cradle rails stop at $X \pm 113$. Outboard pillar centres are $Y \pm 10$ and ± 24 , with 4.5 mm radius, so they clear the wheel envelope. Do not move pillars back beside the axle; that would intersect the 29 mm-wide wheel.

Print sequence and settings

1. Print `fit_coupon.stl`, `motor_cradle.stl`, `arm_base.stl`, `pitch_carrier.stl`, `screen_frame.stl`, `screen_clamp.stl`, `bearing_tower.stl`, `bearing_cap.stl`, and `inner_spacer.stl` first. Use the actual nozzle and filament for the build. The coupon has 3.1/3.2/3.3/3.4 mm bores, 8.4 mm spindle clearance, 22.2 mm bearing clearance and 12.5 mm switch clearance. A 3.4 mm clearance hole must freely pass an M3 screw. A bearing must seat by hand pressure without cracking the tower.
2. Fit the actual motors, servos, board, bearings, switch bushings and connector before committing to the large batch. The cradles use foam and ties, which permit small dimensional differences. Do not scale a whole STL to fix one interface. Change that CAD dimension.
3. Print cosmetic shells in PLA with a 0.4 mm nozzle, 0.2 mm layers and two perimeter lines. These shells are 0.8 mm thick. Keep the designed 2 mm flanges and 2.4 mm seam webs. The geometry provides their thickness; high infill does not strengthen an empty shell. Use three top and bottom layers on flat flanges. Handle shell edges carefully until bolted into rings.
4. Print PETG structural parts with four perimeters, 0.2 mm layers, five top and bottom layers, and 20-25% gyroid infill as the starting profile. Use dry filament. The 5 mm chassis is a connected rib grid with integral bolt bosses. Do not enable vase mode or remove its ribs.
5. Print the chassis, decks, rings, tray, splice plates and bracket bases flat. Print the 95 mm posts upright. Keep the exported orientation for the speaker bracket and screen frame. Add supports under the first head-hub pulley rim, under the arm cups/rods, and inside the dome where the slicer shows unsupported roofs. Remove support from every screw hole and bearing bore.
6. The eye can stand on its rear flange. The hemisphere open face goes on the bed. Its centre boss joins the crown; the centre hole remains open. Print several hemispheres in one batch, without merging them into one object.
7. Use a 5-8 mm brim if a tall part lifts. Inspect Bambu Studio layer previews for two continuous shell lines, retained M3 bosses, complete support contact, and no island warnings. Slice all BOM quantities and record filament mass before continuing. The fit-coupon slice evidence is in `cad/h2d-slice-check.json`; that single coupon does not establish a profile for every part.
8. Paint only after a dry assembly. Keep paint out of bearing bores, nut pockets, shaft fits, threads and belt grooves. Keep the metal-coloured finish light. Primer and filler can add substantial mass.

Assemble the rolling base

1. Put four `chassis_quarter` parts on a flat surface. They have asymmetric mounting holes. Flip the flat plates to obtain the four mirrored quadrants. Do not simply rotate each copy 90 degrees. The finished motor-hole pattern must be $X \pm 79 \pm 121$ and $Y \pm 10 \pm 24$. The battery holes must be $X \pm 45$, $Y \pm 30$, and the electronics-post holes $X \pm 70$, $Y \pm 50$.
2. Fit three `splice` plates under each radial seam, at radii 30, 75 and 150. Each plate straddles a seam. Its two holes lie 8 mm on either side of that seam. Install 24 M3x16 screws, washers and nuts. Leave them loose until the four plate centres meet without a step. Tighten just enough to close the joints.

3. Place one TT motor in each `motor_cradle`, with its long body along Y and the wheel axle pointing outward along X. The yellow gearbox is at the outer fore/aft end. For the rear pair, the axle is toward+Y; for the front pair, toward-Y. The motor body sits on the 3 mm floor. Use thin foam pads on the narrow sides. Start with a 1.7 mm rear stop pad to put the axle near local Y25; verify with the actual drawing and wheel.
4. Put two cable ties through the paired floor slots at local Y-22 and 0. Pass them over the body. Keep them off the axle, motor terminals and gearbox screws. Tighten until the body cannot slide or roll. The ties supply retention; the cradle floor and pillars supply the weight path.
5. Bolt each cradle below the chassis with four M3x65 screws through the 55 mm pillars. Fit a washer under each head and nut. The pillars end at Z72. All four motors must hang to the same height.
6. Press each 3766 wheel squarely onto its TT axle. Hold the gearbox while pressing; do not push against the printed shell. Do not hammer. Verify equal insertion on each side, at least 2 mm clearance from the printed rails, and free hand rotation. The nominal wheel centres are Z31.5, and the 63 mm wheels must clear the chassis at Z72 by 9 mm.
7. Assemble four `bumper_quarter` parts into a ring with eight M3x12 seam screws. Leave the ring off until the electrical drive test has passed. The bottom of the finished bumper is Z15. It must never drag on a flat floor.

Fit the battery and electronics

1. Bolt `battery_tray` to the chassis using four M3x16 screws at X±45, Y±30. Put the heads inside the tray and nuts below the chassis. Cover the screw heads with thin nonconductive padding before placing the battery. Keep the connector facing the rear opening.
2. Fit two 20 mm hook-and-loop straps through the tray. They must independently restrain the battery. A gentle hand shake must not move the battery or pull on its connector. Keep connector strain off the pack leads. Remove the pack for charging and transport.
3. Install four `deck_post` parts at X±70, Y±50. Put `electronics_deck` on the posts, with its speaker-foot holes toward the front. Fit four M3x110 through bolts, washers and nuts. The deck top must be Z175. The battery top remains below this deck.
4. Mount each board on 6 mm `pcb_standoff` parts using the slotted deck. Use actual board mounting holes. The slot grid accepts M2.5 hardware without assuming that every breakout has the same hole pattern. If a board has no mounting holes, use two ties through adjacent slots with nonconductive padding. Do not clamp an inductor, regulator, crystal or connector. Keep the two motor regulators and drivers separated so air can reach them.
5. Fit `speaker_mount` at the front of the deck. Its horizontal foot reaches inward to Y-50 and its two foot bolts align with deck X±30, Y-50. The speaker plate stands at Y-65, from Z175 to 265. Fit the speaker with four M3x12 bolts on its 60 mm mounting pattern. The cone faces forward. Add a thin foam gasket; do not cover the cone.
6. Route cables against fixed walls. Fit strain relief at every moving servo cable. Keep all wiring below the neck bearing deck and out of the belt. Complete the electrical checks in the main assembly guide before installing the shell.

Assemble the skirt and body

1. Insert one standard M3 nut into each hemisphere bottom pocket. Fix each cap with one M3x8 screw and a washer from inside its skirt quarter. Each quarter has six caps. The lower and upper rows sit 23 and 67 mm above that tier's bottom. The cap axis tilts upward with the cone normal: 21.25 degrees on the lower skirt and 15.52 degrees on the upper skirt. The printed mounting bores already use those angles. Do not force the cap to sit radially horizontal.

2. Join four lower quarters with eight M3x12 seam screws. Join four upper quarters with eight more. Fit washers to spread the load on thin webs. Hold the captive nut with a driver while tightening; never twist the shell to react screw torque.
3. Sandwich the chassis between the bumper's top flanges and the lower skirt's bottom flanges. Install 12 M3x16 bolts on radius 170 at angles 15, 45, 75 degrees in each quadrant. The stack is 2 mm bumper flange, 5 mm chassis and 2 mm skirt flange. These are through bolts with nuts; there are no hidden friction-only joins.
4. Bolt lower to upper skirt with 12 M3x12 screws on radius 135. Fit the shoulder to the upper skirt with 12 M3x12 screws on radius 110. Keep the rear two upper-skirt quarters removable. Remove their seam and flange screws when large service access is needed; no glued seam must block battery removal.
5. Join the shoulder front and rear halves with four M3x12 seam bolts. The flat front has reinforced arm pads. Its full circular transition ledges close the space between the flat face and the skirt/neck. The ledges retain the two front skirt bolts and the front neck bolt.

Install the rear display and switches

1. Fit the original T-Display into the screen_frame pocket from behind, screen outward. The 49 x 23 mm front window overlaps the board edge. The 52.2x26mm rear pocket captures the board plan shape. Orient the USB-C port toward the side notch. Check that both buttons can move and the glass does not carry clamping force.
2. Add small nonconductive foam pads at bare PCB edges on the back. Fit screen_clamp, the rear cover. Its central opening gives component clearance. Use padding only where no component or solder joint is loaded. The board must not rattle, bow or be held by its USB connector.
3. The frame's four corner feet are profiled to the curved outer shoulder. The rear cover posts are trimmed to its inner radius. A thin foam washer at each body contact takes up the remaining curved-wall tolerance. Install four M3x25 bolts through the outer frame, shell and cover, with nuts inside. Tighten evenly. The board and screen are visible from the rear at Z323.
4. Connect a USB-C cable through the side notch with the cassette in place. Confirm that the plug can fully seat and can be removed without forcing the board. Leave a service loop on the internal wiring so the entire cassette can be withdrawn.
5. Fit the two NKK S1A switches in the 12.5 mm rear bores at Z279. Use the supplied panel nuts and anti-rotation hardware. Labels must read MAIN and ACTUATORS OFF as specified in the electrical guide. Switch bodies and terminals remain clear of the display. Test access before closing the rear half.

Build and calibrate the two arms

1. Set the four MG92B servos to their firmware centre pulse before installing horns. Use the specified level-shifted control signals. Keep the arms off during this first power test.
2. Bolt each arm_base to its shoulder pad with four M3x16 screws. The bracket back face sits flush on Y-102. The horizontal floor points forward. The bases sit at Z270 and X±43.
3. Place one yaw servo upright in each base. Use foam to centre its shaft at X±43, Y-129. The body rests on the bracket floor. Put two ties through the floor slots and around the body. The shaft must point upward. No tie may cross its moving horn.
4. Use the supplied yaw horn. Attach pitch_carrier to it with two M2x10 screws, washers and nuts through opposite radial slots. The slots are centred 8 mm from the shaft. If the supplied horn holes do not align, drill two 2.1 mm holes in that removable plastic horn using the printed carrier as the template. Keep both holes equally spaced from the centre. Do not drill the servo.

5. Secure the horn to the servo with its supplied centre screw, accessed through the 5 mm central hole. At centre, the pitch-carrier upright wall is on the +X side. Set its underside near Z305. Check that it clears the fixed cradle throughout the intended small yaw range.
6. Fit the pitch servo sideways against the carrier's upright wall. Its output shaft points toward -X. Align the output centre 30 mm above the carrier base and on the carrier Y0 line. Use nonconductive foam behind the case and two ties through the upright wall slots. This mount permits body tolerances while fixing the shaft datum. Keep the case firmly seated without loading its output bearing sideways.
7. Attach the supplied pitch horn to `plunger_arm` or `gun_arm` using two M2x10 screws through the opposite slots perpendicular to the arm tube. These slots pass through only the 3 mm disc. The central 5 mm bore provides access to the supplied spline screw. The rods are decorative. The gun-shaped part has no projectile mechanism.
8. Install the arms pointing forward at the centre pulse. Start with ± 2 degrees on each axis and 0.10 Hz circular motion. Increase only to the firmware's intended small range after clearance tests. The two axes use sine and cosine commands with a quarter-cycle phase difference. Their tips trace circles; the shafts do not spin continuously.
9. Turn off drive power and observe at least 20 slow circles. Check arm-to-arm, arm-to-shell, horn-to-bracket and cable clearance. Route each pitch cable in a loose U-shaped loop back to the fixed base. Tie both ends so the flex occurs in the loop. Stop if a servo buzzes continuously or a cable becomes tight.

Build the bearing-supported rotating head

The bearing axial stack is explicit. From bottom to top: M8 bolt head; 8x12x1mm lower shim at Z404-405; lower 608 bearing Z405-412; printed 12 mm inner spacer Z412-424; upper 608 bearing Z424-431; 8x12x1mm upper shim Z431-432; printed head hub Z432-457; head plate Z457-460; standard M8 washer; M8 nyloc nut. The M8x70 bolt must show at least two full threads beyond the nut. Narrow shims touch only the rotating inner races. A standard 16 mm-OD M8 washer must not be substituted inside the 16 mm cap bore.

1. Bolt `neck_adapter` to the shoulder with three M3x12 screws at radius 110 and angles 30, 150, 270 degrees. The larger inner opening faces upward.
2. Stack a `neck_ring`, three 11 mm `neck_post` spacers, the second ring, three more posts, then `neck_deck`. Align the radius 104 holes at the same three angles. Fit three M3x45 through bolts. The adapter starts at Z367 and the neck deck finishes at Z405.
3. Fit the lower 608 into the bearing tower from its bottom. Fit the upper 608 from the top. Put the 12 mm printed inner spacer between their inner races. Bolt the tower to the neck deck with four M3x12 screws at radius 24. The deck retains the lower bearing's outer race.
4. Fit `bearing_cap` using three M3x8 plastic thread-forming screws in the 2.6 mm pilots. The cap retains the upper outer race at Z431. Its centre bore is 16 mm. Do not overtighten or split the thin outer wall.
5. Insert the M8 bolt, lower narrow shim, inner stack, upper narrow shim and hub as listed above. The hub's 12 mm-diameter foot passes through the cap without contact. The hub foot carries the dome load onto the upper inner race. Its pulley groove centre is Z449.5.
6. Attach `head_plate` to the hub with four M3x16 screws at radius 17. Hold nuts underneath the hub flange. Install the M8 washer and nyloc nut above the plate. Tighten only until axial play disappears. Turn the hub by hand: the head must make a full turn freely, while the tower and cap remain stationary. Excess drag means the inner stack, shim or cap is wrong; do not mask it with more servo power.

7. Bolt head_servo_mount to the deck. Its nominal output centre is X72.5, Y0. Use the four slots to adjust the centre distance. Fit the FS5103R body with two ties and foam pads, shaft upward. Align the supplied horn top with the pulley mounting face at Z445. The 37 mm listed body height fits above the cradle's Z408 floor. Adjust the pads to the actual case and horn; verify the groove plane before tightening.
8. Bolt servo_pulley to its supplied horn with two M2x10 screws. The printed central bore and radial slots are through holes. Use the servo's original centre screw, never a printed spline. Fit the 83 mm-ID, 3 mm-cord NBR O-ring around both pulleys. Adjust the cradle until the belt is lightly tensioned and runs in both grooves. The large groove root diameter is 59 mm; the small is 20 mm. Avoid strong belt tension, which loads the servo bearing.
9. Turn the dome hub by hand through two turns. The belt must track centrally without touching the neck posts or wiring. Then run the servo slowly in both directions with the dome absent. The firmware neutral pulse needs individual calibration; a continuous servo has no position reference.
10. Fit eye to the dome's flat front pad with two M3x16 screws and nuts from inside. The eye faces forward. Insert captive nuts in the two lamp bases and fix the lamps to the raised flat dome pads with two M3x16 screws from inside. The lamps are translucent decoration only. No slip ring or head wiring is required.
11. Insert four M3 nuts in the dome flange pockets. Fit the dome to head_plate with four M3x10 screws from below at radius 101. Keep access to these screws so the dome can be removed later. The head plate and dome rotate together around the M8 spindle.
12. Test one full slow turn in each direction. The eye sweeps beyond the body; leave clear space. Confirm that nothing touches the fixed neck, no cable enters the head, and the belt can slip under a gentle restrained-head test without distorting the servo mount. Do not hold a powered head for more than an instant.

Physical acceptance gates

- All STL and envelope checks must pass before slicing. Re-run `C:/Python314/python.exe scripts/export_cad.py --check-only` after any CAD export. To regenerate all parts, run `C:/Python314/python.exe scripts/export_cad.py --jobs 3`. Individual exports have a 180 second timeout and stop on failure.
- Before fitting the shell, run all four wheels on a stand. Verify side polarity, stop response, current limit and that each wheel clears its cradle. Correct a reversed motor in wiring or the documented firmware channel mapping.
- Weigh the loaded robot. At 3 kg or more, do not assume the selected TT motors can turn it on the chosen floor. Reduce print/finish mass or investigate the actual drive load before driving with the shell fitted.
- Start on a level hard floor, at walking-toy speed, with broad arc turns. Do not assume a zero-radius pivot will work. Wide silicone tires scrub during four-wheel steering. Measure current and temperature during repeated starts and turns. Carpet, ramps and thresholds are outside the initial operating envelope.
- The arm tips must clear the shell and each other through the entire allowed cycle. The head must turn 360 degrees without any fixed contact. Recheck fasteners after the first ten minutes of operation.
- With the network interrupted or browser closed, the firmware must stop motion. The operator must be able to reach ACTUATORS OFF from the rear. Complete the electrical and firmware commissioning checklist before a public demonstration.

Maintenance and access

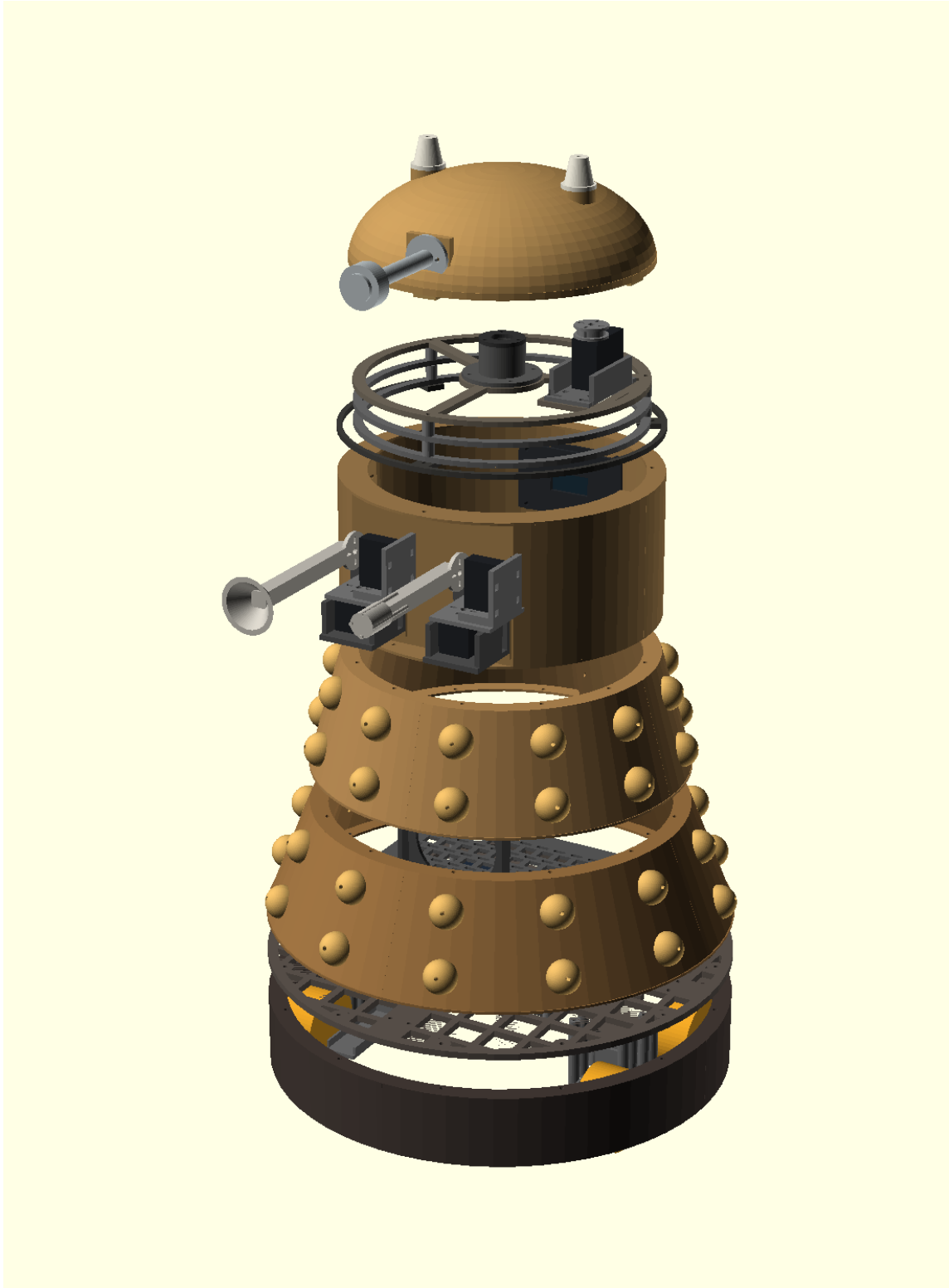
Remove the four display-cassette screws for USB/board access. Remove the shoulder rear half or the two rear upper-skirt quarters for electronics access. Remove the four dome screws for the bearing and belt. Remove the battery from its straps before transport or charging. Keep shell joins bolted; glue is not part of a service joint. Inspect motor ties, belt condition, bearing play and cable loops before every use.

Internal assembly view



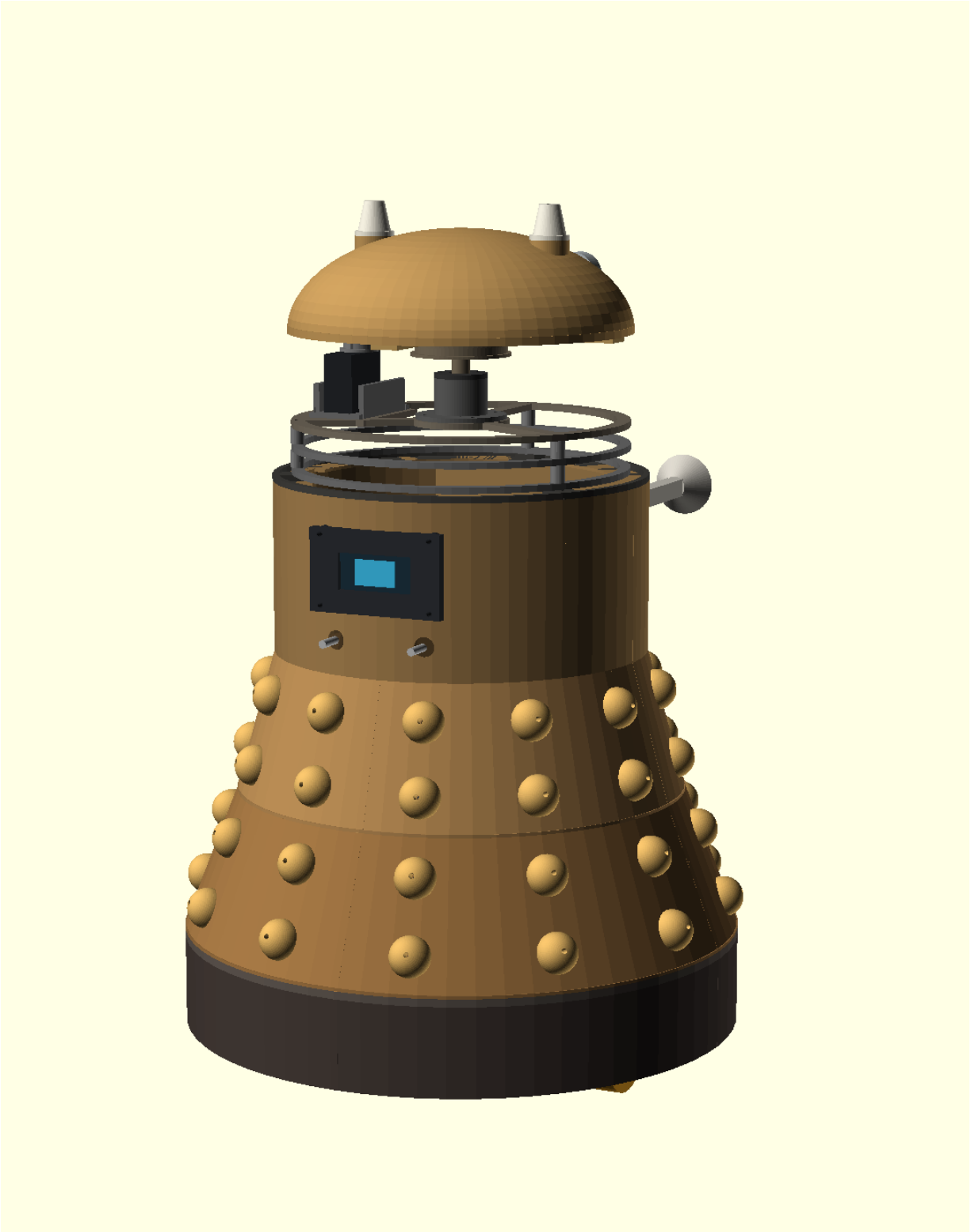
Section through the native CAD. Check battery, drive, deck and rotating-head clearances against the mechanical chapter.

Assembly order view



Separated CAD groups show the shell stack and service joints. Use the part-specific fastener sequence; the gaps in this view are intentional.

Rear display and access



The TTGO display faces outward. Keep the USB opening, operator switches and service fasteners accessible.

Firmware and Wi-Fi controller

This firmware targets the original 1.14-inch LilyGO TTGO T-Display with a classic ESP32, 135 × 240 ST7789 display and 4 MB flash. It does not target the T-Display S3 or the AMOLED version. The display faces outward through the robot's rear hatch. Landscape rotation is 1; use rotation 3 in `src/main.cpp` if the finished mount is reversed.

The software builds for the specified hardware. Its physical operation must be commissioned on the assembled robot. A successful compiler check does not establish stopping distance, servo centre, current draw, Wi-Fi range, speaker level or battery measurement accuracy.

Build and load

1. Install PlatformIO Core or the PlatformIO extension for Visual Studio Code. The pinned packages in `firmware/platformio.ini` install on the first build. Internet access is required for that build only.
2. Turn the physical actuator stop switch OFF. Disconnect the battery before connecting USB. Do not connect two 5 V sources to the board together unless the circuit diagram's power isolation is in place.
3. Use a data-capable USB cable. From the project directory run:

```
Set-Location C:\dev\robots\dalek\firmware

pio run

pio run -t buildfs

pio device list
```

4. Read the ESP32 serial port from the last command. Replace COM7 below with that port:

```
pio run -t upload --upload-port COM7

pio run -t uploadfs --upload-port COM7
```

5. Reset the board. The display must show DISARMED. If automatic flashing does not start, hold the board's BOOT button, start upload and release BOOT after the connection begins. The other button, GPIO35, is the stop/setup button.
6. Disconnect USB before restoring the normal battery supply. Keep the wheels off the floor until all commissioning checks below pass.

There is one factory application partition, 0x10000-0x1ffff (2,031,616 bytes). LittleFS occupies 0x200000-0x3ffff (2,097,152 bytes). Uploading LittleFS replaces the web assets and MP3 files in that partition. Network credentials are in a separate NVS partition. There is no wireless firmware update, cloud service, internet dependency or scheduled background job.

Pinned dependencies: Espressif32 PlatformIO platform 6.5.0, Arduino-ESP32 2.0.14, TFT_eSPI 2.5.43, Adafruit PWM Servo Driver Library 3.0.2, Adafruit BusIO 1.16.1 and ESP8266Audio 1.9.7. The audio dependency is GPL-3.0; retain its notices and supply corresponding source with redistributed firmware binaries. Later ESP8266Audio releases use different ESP32 I2S APIs, so update the platform and audio library together only after testing.

First connection

The board makes a WPA2 access point named DALEK-xxxx. It generates a device-specific 12-character key on first boot and saves it locally. The rear display shows the key. Firmware does not print keys or station passwords to the serial terminal. Join this access point using the displayed key, then open `http://192.168.4.1`. Enter the same key in the control page.

To join a router, open Wi-Fi network settings on the page. Enter the exact network name and its 8-63-byte WPA2 password. This classic ESP32 uses 2.4 GHz Wi-Fi. The password is sent in an authenticated POST request and saved in local NVS. Read the LAN address on the rear display and browse to that address from the same local network. No credentials are present in the repository.

The setup AP remains available while the robot connects to a router. This also provides fallback if the router is missing or its credentials change. Moving between AP and router connections, or losing either connection, disarms motion. Reconnecting never arms the robot. The AP supports two clients, but only one control lease can be active. Keep one control browser open.

Press the rear GPIO35 button to disarm. Hold it for five seconds to erase only the saved router credentials and return to AP operation. The device access key stays the same. The rear key also authorizes robot control for users on the router network. Do not share it widely or expose port 80 to the internet. The local interface uses HTTP; Wi-Fi encryption protects the wireless link, but this is not an end-to-end encrypted service.

Controls

Press Arm controls, then hold a movement button. Forward uses PWM 110/255, reverse 90/255 and turns use 50/255 on the inside wheel pair and 120/255 on the outside pair. Release disarms. Arm again for the next movement. Tight pivots are omitted from the supplied interface because four rubber wheels can scrub heavily against the floor. The requested TT motors have limited torque. Use a smooth, level indoor floor and gentle arcs.

The arm checkbox enables two-axis circles. The radius is 0-12 degrees and frequency is 0.10-0.80 Hz. The default is 8 degrees at 0.40 Hz. One side runs half a cycle behind the other. The head slider commands continuous rotation speed, with zero at the calibrated stop pulse. Hold Arms / head only to animate without wheel movement. These animation settings also apply while a drive button is held. Sliders by themselves do not cause movement.

Speak plays a selected MP3 locally. Silence stops the decoder. Audio operates while the robot is disarmed. All audio stays in the fixed body, so the head can turn indefinitely without twisting an electrical cable. There is no head position sensor or automatic front-facing home position.

The browser must stay visible and focused. Releasing a pointer, losing pointer capture, cancelling a touch, changing tabs, locking the screen, closing the page or pressing Escape requests a stop. Loss of a command response also disarms the interface. Browsers can delay or discard stop packets; the board's independent timeout covers that case.

Electrical contract

Read `electronics-research.md` (`electronics-research.md`) and the power circuit (`../electronics/01-power.svg`), controller, drive and audio wiring (`../electronics/02-controller-drive-audio.svg`), servo wiring (`../electronics/03-servos.svg`) and circuit details (`../electronics/04-circuit-details.svg`) before wiring. All ESP32 signal pins are 3.3 V. Servo signals pass through the specified 74AHCT125 buffers to reach 5 V logic. Never feed 5 V into an ESP32 input.

Function | ESP32 GPIO | Connection

I2C data / clock | 21 / 22 | PCA9685 SDA / SCL, logic VCC 3.3 V

Servo disable | 27 | PCA9685 OE and all used 74AHCT125 active-low output enables

Left forward / reverse | 25 / 26 | Left DRV8833 AIN1+BIN1 / AIN2+BIN2

Right forward / reverse | 32 / 33 | Right DRV8833 AIN1+BIN1 / AIN2+BIN2

Motor enable | 12 | Both DRV8833 SLP pins; 10 kΩ pull-down

Audio bit clock / word clock / data | 17 / 13 / 15 | MAX98357A BCLK / LRC / DIN

Actuator power sense | 36 | 5 V actuator rail through 10 kΩ top / 15 kΩ bottom divider

Battery voltage sense | 39 | Battery through 100 kΩ top / 22 kΩ bottom divider, 100 nF from ADC to ground

Stop / forget router | 35 | Existing rear board button

The PCA9685 address is 0x40. Servo channels are 0 left yaw, 1 left pitch, 2 right yaw, 3 right pitch and 4 continuous head servo. Unused servo connectors remain empty. Each DRV8833 controls one motor on bridge A and one on bridge B. Only its input signals are paired. Never parallel the bridge output terminals.

GPIO27 needs a 10 kΩ pull-up to 3.3 V. Remove the PCA9685 breakout's onboard OE pull-down as specified in the electronics guide; equal opposing 10 kΩ resistors do not produce a reliable HIGH. GPIO12 needs its 10 kΩ pull-down because it is also a boot strap pin. GPIO15 is a boot strap pin connected only to the amplifier's DIN input. Do not add an external pull-up or pull-down there. GPIO36 and GPIO39 are input-only and need the external dividers.

The firmware reserves GPIO5/16/18/19/23/4 for the integrated display. GPIO0 is the existing BOOT button. GPIO34/14 remain reserved for the board's original battery measurement circuitry. The robot's external 12 V pack does not connect to the board's single-cell battery socket.

Stop behavior and limits

The motion task runs independently of the HTTP server and MP3 decoder, with a nominal 10 ms period. It is watched by the ESP32 two-second task watchdog. External enable resistors keep the actuator signals disabled during reset. This is a hobby control system, not a certified safety controller. The physical stop switch cuts actuator power without relying on Wi-Fi or software.

A command lease is granted only after explicit arming. The robot requires a complete valid command at least every 500 ms. The supplied browser sends a fresh command snapshot every 100 ms and aborts a command request after 300 ms. Each arm uses a new random lease; every command needs a strictly increasing sequence number. Old leases and duplicate commands do not refresh the timeout. An expired lease is checked before accepting another command, so a late request cannot revive movement. Invalid control parameters disarm immediately. Software drive values are limited to $\pm 150/255$.

Ordinary wheel speed changes ramp by at most five PWM counts per nominal 10 ms. A requested reversal first ramps to zero, waits 100 ms, then ramps the other way. A stop, fault or disarm immediately drives SLP low and both inputs low; it does not wait for the ramp. This lets the wheels coast. It does not provide a mechanical brake, and stopping distance must be measured with the final mass.

The PCA9685 receives five checked writes every 20 ms. A failed write latches an I2C fault and disables all motion. Fix the wiring and reboot to clear it. Missing PCA9685 at boot prevents arming. No automatic motion recovery follows a power, I2C, Wi-Fi or battery fault. On normal disarm, the arm channels turn off and the head receives its neutral pulse for up to 100 ms before OE disables all pulse outputs. This stops a continuous servo that might otherwise retain its speed after pulse loss. Physical power loss or an I2C fault disables OE immediately. Servo stop behavior and neutral calibration must still be tested. The physical switch is the dependable means to remove torque.

The specified battery is the Bioenno BLF-1206A 12 V nominal 6 Ah LiFePO4 pack. The software multiplies the ADC voltage by $(100 + 22) / 22 = 5.54545$ to recover pack voltage. It smooths readings at 10 Hz. Below 11.2 V for one second, it disarms; voltage must recover to at least 12.0 V before re-arming is allowed. A disconnected ADC or an implausible voltage also prevents arming. These thresholds are conservative operating limits, not a battery

state-of-charge meter, a charger or a replacement for the pack's protection circuit. Calibrate the ADC against a multimeter before use.

Servo and sensor calibration

All calibration values are in `firmware/include/config.h`. Keep the wheels lifted and the servo horns removed for initial checks.

1. Check 5 V rails with actuator power disabled. Verify that neither DRV8833 SLP nor PCA OE can enable motors during ESP32 reset.
2. Measure battery voltage with a multimeter. Read the displayed voltage. Set `PACK_ADC_CORRECTION` to measured voltage divided by displayed voltage, rebuild, and recheck at two battery voltages. Do not lower a cutoff to hide a wiring error.
3. Leave arm and head commands at zero. Enable actuator power, arm, and hold Arms / head only. The four positional servos must settle near their centres. Fit each horn with its joint centred and no mechanical load, then stop before fitting links. The initial centres are 1500 μ s; change `ARM_CENTER_US` in small steps only if needed.
4. With the head drive belt/linkage disconnected, keep head speed zero and hold Arms / head only. If the head servo creeps, adjust its physical neutral trimmer if present. Otherwise adjust `HEAD_NEUTRAL_US` in 5 μ s steps until rotation stops. Disarm between changes. Couple the head only after this check.
5. Start arm circles at radius 2 degrees and 0.10 Hz. Confirm that yaw and pitch are both active. Confirm that neither joint touches a stop. Increase to 8 degrees and 0.40 Hz. The pulse envelope stays within 1200-1800 μ s, but mechanical clearance is the limiting factor. Reduce the maximum radius if the actual links require it.
6. Increase head speed slowly in each direction. `HEAD_MAX_DELTA_US` is 180 μ s; the control maps -100...100 to neutral ± 180 μ s. `HEAD_DIRECTION` reverses the sign. The servo is a speed actuator; the slider is not an angle command.
7. Lift all wheels. A forward command must rotate all four wheels toward forward travel. If one motor is reversed, swap only that motor's two output wires. `LEFT_DIRECTION` and `RIGHT_DIRECTION` reverse an entire side if necessary. Stop and remove actuator power before rewiring.
8. Play a phrase. The fixed digital gain is 0.28, below the design ceiling of 0.35. MAX98357A hardware gain must be 3 dB as specified in the circuit guide. Do not raise gain above 0.35 with the 8 Ω 1 W speaker and specified logic supply. Digital gain is not a calibrated wattmeter; verify clean sound, supply stability and acceptable speaker temperature.

Acceptance checks on the completed robot

Perform these with a helper at the physical stop switch, first with wheels lifted, then at low speed on a clear floor. Record the results in the assembly commissioning sheet.

- Power up, reset and reconnect USB: no wheel or servo motion before an explicit arm command.
- AP mode: router absent, page and every local voice phrase work through 192.168.4.1.
- Station mode: join the private router, read the rear LAN IP, load the page and drive through that address.
- Release the movement button: wheel outputs disable and the page becomes disarmed. Measure actual coasting distance with the final shell and battery fitted.
- While moving, disable the phone's Wi-Fi: commanded drive must cease within the 500 ms lease timeout plus control-task scheduling delay. The robot must remain disarmed after reconnection.

- While moving, lock the phone or change tabs: motion stops. Unlocking does not resume the prior command.
- Press physical stop: motor and servo supply must lose power even if the ESP32 keeps running. Releasing the switch does not restart movement.
- Press the rear GPIO35 button while holding a browser control: motion stops and does not restart without a new arm.
- With actuator power off, disconnect the PCA9685, then boot: I2C fault appears and Arm is refused. Reconnect only with power removed.
- Use an adjustable current-limited bench supply in place of the pack for the voltage check. At 11.0 V for more than one second the software must disarm. At 11.5 V it must refuse a new arm. Above 12.0 V it may accept a new explicit arm.
- Run arm circles and head motion while playing all phrases; no reset, supply sag, decoder stall or joint contact is acceptable.
- Complete gentle forward arcs on the intended floor. Measure individual motor current, case temperature and stopping distance. Do not accept a stalled inside wheel or prolonged current-limit operation.

Automated checks

From firmware/:

```
pio run
pio run -t buildfs
node --check data/app.js
node test/test_ui.js
$env:PATH = 'C:\msys64\mingw64\bin;' + $env:PATH
g++ -std=c++11 -Wall -Wextra -Werror -I include test/test_control.cpp -o $env:TEMP\dalek-control-test.exe
& $env:TEMP\dalek-control-test.exe
```

The host C++ test runs the same control-state code used by the ESP32. It checks strict input parsing, boot lockout, expired and replayed leases, reconnect behavior, unhealthy-input disarm, millisecond rollover and reversal delay. The Node test exercises the actual browser script with a simulated DOM and transport. It does not replace a real browser or robot commissioning test.

Sources

- [LilyGO original T-Display board, schematic links and pin allocation.](#)
- [PlatformIO LilyGo T-Display board definition.](#)
- [TFT_eSPI original T-Display setup.](#)
- [Adafruit PCA9685 driver source and checked setPWM result.](#)
- [ESP8266Audio version 1.9.7 source and license.](#)
- [ESP8266Audio current API compatibility notes.](#)

Dalek electronics design and source record

Research date: 2026-09-11. This is a build design, not a claim that a robot was physically assembled or tested. Listed prices are USD before shipping and tax. Stock is a page observation, not an order reservation. The bill of materials separates checked prices from allowances.

Selected build

The controller is the original 1.14-inch TTGO T-Display ESP32. It is mounted on the rear with its screen visible. Four Adafruit 3777 motors each drive one Adafruit 3766 wheel. Each motor has a separate current-limited H-bridge. Two positional servos move each arm in yaw and pitch. Their phased movement draws a circle with the arm tip. A continuous rotation servo drives the head through the mechanical bearing and belt. No wires enter the rotating head.

Power comes from one removable Bioenno BLF-1206A 12 V 6 Ah LiFePO4 pack. Three independent converters produce 5 V for drive motors, 5 V for servos, and 5 V for controller/audio. The physical ACTUATORS OFF switch removes input power from both motion converters. Logic and the display remain on. All circuit grounds join at a power distribution star.

The authoritative connection set is `electronics/wiring.csv` together with the circuit SVGs. Use the component references from those files. Pin numbers on the TTGO are GPIO numbers. Pin numbers on U8/U9 are DIP package pin numbers.

Required motors and wheels

Adafruit 3777 accepts 3-6 V. The vendor measured 160 mA unloaded and 1.5 A stalled at 6 V on a sample. The listing gives 0.8 kg·cm stall torque at 6 V, a 48:1 gearbox, 70×22×18 mm body and 30.6 g mass. There is no encoder. The selected 5 V rail stays within the motor voltage range. Four motors cost \$11.80 at the checked \$2.95 each. These numbers do not establish continuous load torque. [Adafruit 3777](#)

The required orange/clear 3766 is a press-fit TT wheel, 63 mm diameter by 29 mm wide, 38 g. The checked price is \$1.50 each. Adafruit marked it out of stock. This design retains the user's exact wheel. Obtain the wheels before accepting the final motor-to-shell fit. Do not buy a different wheel from a similar photograph. [Adafruit 3766](#)

A nominally stronger DRV8871 was rejected because its minimum motor supply is 6.5 V. That is above the required motor's 6 V maximum. Feeding the motors an overvoltage and merely lowering PWM duty is not used here. [Adafruit DRV8871](#)

The selected two DRV8833 boards each contain two bridges. They support the 5 V motor supply. Each stock 0.2 ohm sense resistor sets approximately 1 A limit through 0.2 V/R. Keep the factory resistors and leave their bypass jumpers open. Tie AIN1 to BIN1 and AIN2 to BIN2 on each side. Do not tie the bridge outputs together. This gives the controller one forward/reverse input pair per side while each motor has its own current limit. SLP is active high, with an internal weak pull-down. Add the specified external 10k pull-down. [DRV8833 product](#), [DRV8833 pinouts](#)

The four-wheel skid-steer layout has no precise speed or distance measurement. The body must be light. Three kilograms is a design ceiling to test, not a proven payload. Use hard flat indoor flooring. Wide silicone tires can resist turning. A short wheelbase reduces this resistance. Do not use the robot on stairs, ramps, carpet, or outdoors.

Servos and movement

The four arm servos are MG92B units, Adafruit 2307. The listing specifies 5 V signals as well as 5 V power. Direct 3.3 V PWM is therefore not the chosen wiring. The listed torque is 3.1 kg·cm at 4.8 V and 3.5 kg·cm at 6 V; weight 14 g; envelope 36×12×31 mm; 20-tooth horn spline. The listing also says this servo has no mechanical end stops. Use its supplied horn rather than printing a nominal spline. Begin with a small software travel window and check every linkage. [MG92B](#)

The head uses the FS5103R, Adafruit 154, which controls speed and direction rather than shaft angle. The source lists 4.8-6 V power, 3 kg·cm stall torque at 4.8 V, 40 g weight and 25-tooth horn spline. It is currently out of stock. Its source contradicts itself about the calibration screw and explicitly notes that newer units lack the adjustment hole. Set neutral in firmware; do not instruct the builder to drill or open the servo. The bearing carries the head mass. [FS5103R](#)

U10 is Adafruit 815 PCA9685. Use 3.3 V VCC and its default I2C address 0x40. The board's I2C pull-ups then stay at the ESP32 logic voltage. Leave the V+ rail disconnected because servo power is distributed separately. The board's output-enable pin is active low and normally pulled down. Remove that pull-down resistor, then fit an external 10k resistor from OE to 3.3 V. Identify it by continuity to OE and GND against the actual PCB revision; do not remove a resistor by guessed reference number. Measure the result before connecting servos. [PCA9685 pinouts](#)

U8/U9 are 74AHCT125 buffers. Power them from 5V_SERVO. Their inputs recognize the 3.3 V PWM; outputs provide 5 V pulses. Use the AHCT version, not HC. Verify the actual purchased chip marking and its data sheet. The TI SN74AHCT125 specifies input leakage at VCC=0 through 5.5 V and input thresholds compatible with 3.3 V. This matters because the controller remains on when servo power is disconnected. The buffer input must not power its dead supply through a positive clamp diode. Every used buffer OE joins SERVO_DISABLE. Every unused input has an explicit tie. Two 100 nF capacitors sit at the IC supply pins. [Adafruit 1787](#), [TI SN74AHCT125 data sheet](#)

The channel order is 0 left yaw, 1 left pitch, 2 right yaw, 3 right pitch, 4 head speed. Use 50 Hz. With horns removed, start at 1500 μ s. Fit arm horns at the mechanical center. Start circular motion with a 2-degree angular radius at 0.10 Hz. Increase only after checking clearance. The normal default is 8 degrees at 0.40 Hz. Phase the pitch 90 degrees from yaw. This is a conical arm-tip path, not continuous rotation of the entire arm shaft. Head neutral is calibrated separately. Continuous rotation cannot command an absolute head angle without feedback, which this design does not include.

Power and battery

The selected BLF-1206A provides 12 V nominal, 6 Ah, 72 Wh by nominal multiplication, 12 A continuous discharge, and internal protection and cell balancing. The source gives 0.7 kg mass. Its metric and inch dimensions differ: 108×64×69 mm versus 4.4×2.6×2.8 in, which converts to 111.8×66.0×71.1 mm. The design uses the larger converted dimensions and a minimum 116×71×76 mm padded cavity, plus cable clearance. It uses the factory PP30 discharge connector. Never open or modify the pack. [Bioenno BLF-1206A](#)

Use only the matched BPC-1502DC charger for this build: 14.6 V, 2 A with the specified DC charge plug. Switch off, unplug the PP30 discharge connection, and remove the pack from the printed shell before charging. Charge on a nonflammable surface with the pack and charger visible. Do not charge through TTGO's lithium battery connector. This robot has no onboard charging circuit. [Bioenno matched charger](#)

U2/U3 are Pololu 4091 D36V50F5 regulators. Their fixed 5 V outputs, input range and current capability give substantially more supply margin than the rejected low-voltage pack arrangement. The manufacturer's output-current rating depends on input voltage and temperature. The 5.5 A name is not an unconditional guarantee inside a printed enclosure. Leave the boards ventilated and solder thick power conductors directly to their

duplicated power pads. Do not send more than 3 A through one ordinary header pin. The board is 25.4×25.4×9.5 mm and has M2 mounting holes. Leave EN, PG and VRP unconnected. [Pololu 4091](#)

U4 is the Adafruit 4739 MPM3610 fixed 5 V regulator. It is rated for 1.2 A and up to 21 V input. The checked price is \$6.95. This lower-power converter is sufficient only with the specified audio limit. CAD from the manufacturer's board repository gives 10.16×17.145 mm outline and a 2.5 mm mount hole at (5.08, 14.605) mm. The 5 V variant uses the same outline; verify the actual board before fastening. [MPM3610 product](#), [Adafruit board CAD](#)

Use NKK S1A switches for MAIN and ACTUATORS OFF. Their source rating is 20 A at 30 VDC; the panel opening is 12.5 mm. ON joins terminals 1 and 3. The small Adafruit covered toggle was rejected because Adafruit recommends only 1-2 A despite the 20 A marking. Each battery/branch fuse has a stated value. The main 7.5 A fuse protects the harness close to the battery. Branch fuses are 3 A at each motion converter input and 1 A at the logic converter input. These are wire-fault protection, not a precision current limiter or motor thermal model. [NKK S1A](#), [Adafruit covered toggle](#)

Current allowances at 5 V are 4 A drive, 5 A servos and 0.9 A logic/audio. Total peak output allowance is 49.5 W. At 11.2 V battery and an assumed 85% conversion efficiency, estimated input current is 5.20 A. This leaves margin below the 12 A battery limit. The 5 A servo allowance is a design allowance because the vendor pages do not provide an authoritative stall-current value. Measure the delivered units. Sustained stall of all servos is not an operating condition. If a rail falls below 4.8 V, a fuse opens, or a driver repeatedly shuts down, stop and remove the mechanical obstruction; do not increase fuse or sense-current limits.

The 100 ohm 1 W discharge resistors each consume 0.25 W. They reduce residual actuator voltage after S2 opens. Capacitors still contain energy for a fraction of a second. The disconnect causes coasting; it is not an instantaneous mechanical brake or a certified emergency-stop system. Prove that both actuator rails fall below 0.5 V within 1 second during the commissioning test, including the no-load case.

Controller and Wi-Fi connections

The original board uses ST7789 display pins 19, 18, 5, 16, 23 and 4. Buttons use 0 and 35. Its onboard single-cell battery measurement is GPIO34 with GPIO14 power control; that circuit must not measure the 12 V pack. The chosen separate ADC is GPIO39(SVN). GPIO36(SVP) reads actuator power state. Both are input-only. The pin map source confirms those pins are exposed. [LILYGO pin map and source](#), [manufacturer pin-map image](#)

External assignments are I2C 21/22, servo-disable 27, motor-enable 12, left drive 25/26, right drive 32/33, audio 17/13/15, battery 39 and actuator sense 36. GPIO12 has a 10k pull-down to keep the normal boot strap state. GPIO15 connects only to the amplifier input without added bias. No GPIO receives 5 V. Use 2.4 GHz Wi-Fi. The firmware provides its own access point and a configuration page to join an existing network. Keep the rear antenna area clear of the battery, motor cans, foil paint and metal fasteners; use an uncoated plastic region around it.

The pack divider is 100k top/22k bottom, 1% resistors, with 100 nF at GPIO39. At 14.6 V it gives 2.633 V. Multiplier is 122/22. A 100k/33k divider is NOT used; it would exceed the intended ADC range on this pack. A second 10k/15k divider maps 5V_MOTOR to 3.0 V at GPIO36, with 100 nF filtering. Firmware stops on a sustained 11.2 V reading and permits re-arm only above 12.0 V. LiFePO4 voltage does not provide a reliable linear charge percentage. Display voltage or a conservative low-battery state, not a claimed precise fuel gauge.

Sound

The ESP32 decodes the supplied MP3 files and sends I2S to Adafruit 3006 MAX98357A. It needs BCLK on GPIO17, LRC on GPIO13 and DIN on GPIO15. It does not require MCLK. OUT+ and OUT- go only to the speaker. Neither speaker lead connects to ground. Use a 100k resistor from GAIN to 5V_LOGIC for 3 dB gain, and leave SD at its factory mono-mix configuration. [MAX98357 pinouts](#)

The speaker is Adafruit 1313, 8 ohm 1 W, 77.8 × 77.8 × 25.49 mm. Its four mounting tabs are 60 mm apart. It costs \$1.95 and was in stock. The supplied firmware uses fixed digital gain 0.28. This is a design setting, not an acoustic measurement. Verify that continuous test audio is below 2.83 Vrms across the 8 ohm load, using a suitable differential measurement and audio-band filtering. Never attach a grounded oscilloscope clip to an amplifier output. Normal spoken clips have lower average power than a continuous test tone. [Adafruit 1313](#)

Harness assembly

1. Disconnect the battery. Fit the switches, fuse holders and ground-star block. Label every rail and both switch OFF positions. Protect all exposed lugs with heat shrink.
2. Fit U2/U3 on insulated standoffs with air space around them. Fit U4 and the two prototype boards. Fit U1 in the rear holder with the display facing outward. Keep the USB connector accessible.
3. Assemble the buffer board without ICs installed. Use DIP14 sockets. Check pin 14 to 5V_SERVO and pin 7 to GND. Check every channel against sheet 03. Add one 100 nF capacitor per socket. Fit the input and output pull-downs and 220 ohm output resistors.
4. Modify PCA OE as described above. With the board unpowered and isolated, check that OE is not still tied to ground through its original resistor. Add the external 10k to 3.3 V. Record the actual board revision and removed resistor location for later service.
5. Wire the power branches in 18 AWG. Wire one 22 AWG positive/ground pair from the servo distribution block to each servo plug. Do not route aggregate servo current through the PCA header rail or prototype-board strips.
6. Wire each motor to its own bridge. Add 100 nF directly at each motor's solder tabs. Retain only short lengths of the supplied 28 AWG leads. Twist the two motor conductors. Add strain relief near the plastic gearbox, not on the motor tabs.
7. Wire signals from the CSV. Keep I2C and I2S away from motor wiring. Add service loops at the removable rear hatch. Keep wires out of belts, wheels and arm linkages. The head must have no wiring.
8. Fit and label the removable J_USB_ISO run-power connection. To program: switch MAIN off, unplug the battery, open the run-power link, then attach USB. Unplug USB before reconnecting the link and battery. No external battery is connected to the TTGO JST socket.

Bench acceptance before the shell is closed

Use a current-limited bench supply for first power where available. Otherwise leave the motors and servos unplugged until the polarity and rail checks pass. Keep the robot raised so all wheels are clear. Keep hands clear of the arm and belt paths.

1. With no power, check for shorts between each positive rail and GND. Check for no direct connection between the three 5 V outputs. Check capacitor polarity and fuse values. Check each power connection by meter, not wire color alone.
2. MAIN on and ACTUATORS off: measure logic 5 V; verify rear screen boots. Verify both actuator rails discharge below 0.5 V within 1 second. Check no servo power appears through a signal wire.
3. ACTUATORS on but disarmed: measure all three rails 4.8-5.2 V. Confirm the battery reading agrees with a meter within 0.2 V after calibration. Verify GPIO36 is about 3.0 V.
4. Hold the TTGO reset button. Verify motor SLP below 0.3 V and PCA OE above 3.0 V. Motors must not run. Release reset; robot stays disarmed. Power-cycle several times with ACTUATORS on and watch for unintended movement.

5. Fit one motor at a time. At low duty, verify direction and the commanded stop. Do not deliberately stall a motor. Then connect all four and repeat with wheels raised.
6. Fit servos with horns removed. Verify neutral and channel mapping. Fit horns at the mechanical center, then make slow small circles. Check every linkage and head belt for binding. Set the head neutral until it stops without creep.
7. While moving at low speed, close the browser, turn off Wi-Fi and disconnect the control device in separate tests. The firmware timeout must stop all motion. It sends a calibrated head-neutral pulse for 100 ms before disabling servo outputs. Reconnection must require manual arming. Turn ACTUATORS off and verify physical power removal while the display remains on.
8. Simulate low battery with an adjustable supply rather than over-discharging the pack. Confirm sustained 11.2 V causes stop, 12.0 V is required before re-arm, and an invalid ADC prevents arming.
9. Weigh the finished robot. First drive on a hard flat floor at low duty. Test straight motion and broad forward turns. The supplied control interface intentionally uses arcs rather than in-place turns. Log loaded pack voltage, each 5 V rail minimum and motor-driver temperature. No rail may drop below 4.8 V; there must be no reset or repeated thermal shutdown. Stop if the robot binds while turning. A smaller/lighter shell or lower-friction floor is required if the specified motors cannot turn it reliably.
10. Run 15 minutes of representative driving and animation with pauses. Confirm regulator and driver package cases remain below 60°C with a thermometer. Motors, servos and printed mounts must stay below the assembly guide's more conservative 50°C limit. Wire joints must remain cool and wheels must not rub. These are project acceptance limits, not claimed vendor maximums. Test audio with drive and arms active. Check there is no reboot or audible power noise.

Calculations and limits

electronics/calculations.json is generated from the documented assumptions by python electronics/generate.py. At the 1 A current limit, a simple linear interpolation from the vendor's 6 V current/torque data estimates 0.0492 N·m per motor, or 6.25 N combined wheel force at 31.5 mm radius. Gearbox losses, current ripple and unit variation reduce confidence in this estimate. It is not a continuous-torque rating. The unloaded 6 V, 250 RPM sample would imply 0.825 m/s wheel speed; the actual 5 V loaded robot will differ. PWM limits are not a speed governor without encoders.

With 75% of nominal 72 Wh treated as useful energy, runtime estimates are 3.6 hours at 15 W average and 2.16 hours at 25 W average. These are planning values. The builder must measure loaded current and an actual discharge cycle to state runtime. Do not describe the result as verified until the physical checks above have passed.

Open procurement items are the required 3766 wheels and selected 154 head servo, both marked out of stock. The two Pololu regulators are rationed with backorders allowed. The pin assignment, circuit files and firmware can be checked now. Hardware fit, turning ability, noise, servo current and runtime still require the physical build.

Design research and engineering decisions

This package describes a small, original Dalek-shaped indoor robot. The design uses the requested Adafruit TT motors and orange wheels, an original TTGO T-Display ESP32, a printed shell, two moving arms, and a head that can turn through repeated full revolutions. It is a fabrication design for a first prototype. The CAD and software can be checked digitally. Their operation under actual load still needs the commissioning tests in `assembly.md`.

Source method and date

Product facts were checked against manufacturer product pages, dimensional drawings, component datasheets, and official board source files on 11 September 2026. The required motor and wheel pages were checked first. Component research and exact links are also recorded in `electronics-research.md`, `mechanical.md`, and the BOM CSV files. Prices exclude shipping, tax and regional differences. A product being listed does not guarantee stock when an order is placed.

Quoted electrical ratings describe a specific part and operating condition. Design targets, calculated limits, and assumptions below are identified separately. Neither a successful firmware compile nor a closed STL mesh establishes that a physical robot is safe to drive.

Size and shape

The final CAD is 541 mm tall, or about 21.3 inches, comfortably below 914.4 mm, or three feet. Its base is 360 mm in diameter. It uses a broad skirt, rows of raised hemispheres, a narrower shoulder section, neck rings, a dome, an eye stalk, a plunger-shaped arm and a blunt decorative emitter arm. These parts form an original interpretation of the character rather than a dimensionally exact replica of a television prop. No downloaded prop mesh or prop plan is redistributed.

The shell is split into parts because the assembled footprint is wider than the H2D's single-nozzle travel. Bambu specifies 325 x 320 x 325 mm for single-nozzle printing and 300 x 320 x 325 mm for dual-nozzle printing. The often-advertised 350 mm width is the combined reach of two nozzles; it is not a 350 mm wide region available to either individual nozzle. The generated print orientations are checked against a conservative 300 mm cube. Use the supplied print manifest rather than scaling an STL to make it fit. Scaling would also change bearing seats, motor pockets, holes and screw spacing. [1]

PLA is appropriate for cosmetic panels used indoors. PETG is preferred for load-carrying mounts, clamps and parts near motors. Neither material should be assumed to tolerate the inside of a hot parked car. Printed walls, service openings and internal fasteners have more effect on fit and durability than cosmetic surface smoothing. Print the fit parts before spending material on the full shell.

The requested drive parts set the operating limit

Adafruit 3777 is a low-cost 1:48 plastic gearbox motor. Its allowed supply is 3-6 V. Adafruit reports 1.5 A stalled at 6 V and 0.8 kg.cm stall torque at that voltage. Adafruit 3766 is a 63 mm diameter, 29 mm wide press-fit wheel for the TT shaft. Four separate motors drive four wheels. The wheel page was out of stock when checked; the design retains that exact requested wheel. [2,3]

Stall torque is a brief limiting value, not a continuous operating rating. At the 6 V specification, 0.8 kg.cm is approximately 0.0785 N.m. With a 31.5 mm wheel radius, four motors give an ideal combined stall force of $4 \times 0.0785 / 0.0315 = 9.97$ N. At the chosen 5 V supply and with current limiting, the available force is lower. Gear loss, shaft friction, unequal load and warm windings lower it further. This calculation is an upper bound; it is not a payload rating.

For a design example, a 3 kg robot on a smooth floor with an assumed rolling resistance coefficient of 0.03 requires about 0.88 N just to roll. An acceleration of 0.15 m/s² adds 0.45 N. That 1.33 N example is plausible for gentle straight motion but does not include steering scrub. The coefficients and acceleration are engineering assumptions, not measurements of the supplied wheels. Carpet, door thresholds and slopes are outside the initial operating envelope.

With four fixed wheels, steering requires the tires to slide sideways. For an example friction coefficient of 0.6, mass 3 kg and 130 mm wheelbase, a rough upper bound for stationary scrub resistance is $0.6 \times 3 \times 9.81 \times 0.130 / 2 = 1.15$ N.m. The ideal differential drive torque at the 6 V stall specification with a 260 mm track is about 1.30 N.m. The small margin disappears at 5 V and reduced current. A stationary pivot therefore cannot be promised on a high-grip floor. Use broad moving arcs first; release the control immediately if a wheel stops. The comparatively short wheelbase reduces this resistance while the wide track supports stability.

Motor speed is also open loop. The motors have no encoders. A PWM percentage controls electrical drive, not a measured speed in metres per second. Motors may need polarity and trim checks. The firmware ramps drive commands and limits their range. Wheel diameter and unloaded motor speed can estimate an upper speed, but loaded speed and stopping distance must be measured on the completed robot. [2]

Driver selection

Two Adafruit DRV8833 boards provide four H-bridges, one bridge for each motor. Each pair of motors on a side receives the same logical direction command. The motor outputs are not wired together. The Adafruit breakout has a nominal 1 A current limit per bridge. Retain its current-sense resistors. A current limit reduces damage during a fault but does not make a stalled motor acceptable for continuous operation. [4]

The DRV8833 operating supply range includes 5 V. The TI datasheet distinguishes continuous, peak and package-dependent current values. The actual breakout's default limit and heat dissipation govern this build. It is not valid to choose a driver merely because a catalogue headline says 2 A peak. Reset-state wiring also matters: the shared sleep line must be low at ESP32 reset, not left to firmware that has not yet started. [4,5]

Battery and power distribution

The selected pack is the Bioenno BLF-1206A, a protected 12 V, 6 Ah LiFePO₄ pack with a stated 72 Wh energy rating. The manufacturer lists a 12 A continuous discharge limit, approximately 0.7 kg mass, and the BPC-1502DC 14.6 V, 2 A charger. Its internal protection and large energy reserve are useful for a mobile prop. It supplies converters; it must never connect directly to a 5 V rail, a TT motor, or the T-Display battery socket. [6]

This is the main deliberate exception to an Adafruit-only shopping list. Adafruit's 3.7 V, 6600 mAh pack has a much smaller energy reserve and a stated 3.3 A maximum discharge. Its own page recommends a substantially lower sustained draw. Boosting its output to 5 V while driving four motors and five servos would demand more battery current than the output current. Capacity in mAh alone is not a sufficient selection rule. A protected ready-built pack avoids assembling and balancing loose cells. [7]

Separate regulated 5 V supplies feed the wheel motors, servos, and logic/audio. This prevents a servo pulse or motor start from drawing through the ESP32 board's small power path. All returns meet at the common ground distribution point. The physical actuator switch removes energy from both movement supplies while leaving the controller display available. The main fuse is close to the battery. Wire size, connector ratings, branch fuses and the actual switch DC rating appear in the electrical package.

Runtime is an estimate until measured. With 72 Wh nominal and an assumed 75% usable energy allowance, 54 Wh remain. At an estimated average load of 15 W, that is 3.6 hours; at 25 W, 2.16 hours. These values cover energy only. Temperature, battery condition, low-voltage shutdown, repeated starts and actual servo load change the result. Do not call any of these a guaranteed runtime. Record the first full discharge test with voltage, duration,

activity and temperature. [6]

Arms and head

Each arm has two perpendicular positional servo axes. The controller varies the two angles using sine and cosine signals. For small angles, the arm tip follows a near-circular orbit. At large angles, the geometry becomes a spatial curve, so the firmware uses a modest amplitude. Equal pulse amplitudes do not guarantee equal physical angles until the servo centres and directions have been calibrated. The two assemblies can share a phase offset without needing a cam mechanism.

Four MG92B servos provide the arm axes. Their power and signal requirements must both be met: the Adafruit product page specifically calls for 5 V control signals. A PCA9685 running at 3.3 V does not provide a 5 V HIGH just because its separate servo power terminal has 5 V. Two AHCT buffers translate the five control signals. The rear electronics drawing shows their enable pins, power bypassing, and unused inputs. [8]

The head uses an FS5103R continuous-rotation servo to drive a bearing-supported rotating assembly. The servo controls speed and direction, not an absolute compass angle. Neutral pulse width needs calibration to stop creep. A continuous servo cannot provide a reliable “face forward” command without a separate reference sensor; no such positioning claim is made. The bearing carries the dome load. The servo supplies rotation rather than supporting the full head as a cantilever. [9]

The eye and dome decorations contain no moving electrical wires. Audio comes from a speaker fixed in the body. This permits repeated complete rotations without a slip ring. It also removes rotating contacts and their noise from the audio power path. All belt, bearing, shaft, retention and horn hardware is enumerated in the mechanical bill of materials.

Controller and network

The design is for the original ESP32 TTGO T-Display with a 1.14-inch ST7789 display and 4 MB flash. LilyGO publishes the TFT pins, button pins, battery-monitor pins and board drawings. The S3 and other T-Display models have different dimensions and pin assignments and cannot use this firmware and rear mount unchanged. The screen faces outward at the back. Leave its antenna area clear of wires, metal tape and the battery. [10]

ESP32 Wi-Fi supports station mode, access-point mode and their combined use. The robot serves its own browser interface. A nearby phone can connect directly without an Internet connection, or the robot can join an existing 2.4 GHz network. Station credentials are entered locally. No cloud broker or paid service is needed. The exact setup, fallback behavior and displayed addresses are described in `firmware.md`. [11]

Motion needs a current command stream and an explicit arm action. A lost connection, stale command sequence, browser release, physical actuator power loss or safety fault must remove drive commands. Software timeout is an additional control, not a substitute for the physical power switch. The network is local and HTTP is unencrypted; do not expose it through router port forwarding. The Wi-Fi and application password must remain private to the operator.

Voice collection

The package includes 24 newly synthesized MP3s, about 99 seconds total. Their measured file sizes and durations are recorded in `audio/catalog.json`. They use original short transcripts, the installed desktop speech engine, and a reproducible metallic voice effect. They do not contain ripped television clips. The source list is editable and the generation scripts are supplied. All clips are stored in ESP32 flash, so a memory card or network download is unnecessary.

A MAX98357A converts digital audio to speaker power. The selected 8 ohm, 1 W speaker and limited playback gain keep demand within the logic supply budget. The two speaker terminals form a bridge output: neither is a ground terminal. Fix the speaker to the stationary body and route its wires away from the analog voltage-sense leads. Refer to the electrical source notes for amplifier gain and speaker limit calculations.

What is established, and what must be measured

Digital checks establish that the delivered files exist, mesh volumes are positive, exported parts fit the selected print envelope, audio decodes, and firmware compiles. Inspection checks that connector names and pins agree across code and diagrams. These checks are valuable but cannot detect actual servo backlash, moulding variation, printer shrinkage, real traction, bearing friction, battery noise or a loose screw.

The first build must measure fit, loaded mass, wheel clearance, free head travel, arm current, supply sag, temperature and stopping behavior. The assembly guide makes those checks before closing the shell and before the first floor run. If the requested drive parts cannot move the actual mass on the actual surface, the correct response is to reduce mass or change the operating surface. Do not remove current protection or raise motor voltage above its rating to force a result.

References

1. Bambu Lab. H2D technical specifications, single/dual nozzle build volumes.
https://cdn1.bambulab.com/documentation/h2d/en/H2D_Laser_Full_Combo_20250305.pdf and
<https://eu.store.bambulab.com/products/h2d> . Manufacturer PDF indexed text and regional product specifications checked 2026-09-11; direct fetch intermittently unavailable.
2. Adafruit Industries. DC Gearbox Motor - TT Motor, product 3777. <https://www.adafruit.com/product/3777> .
Checked 2026-09-11.
3. Adafruit Industries. Orange and Clear TT Motor Wheel, product 3766. <https://www.adafruit.com/product/3766> .
Checked 2026-09-11.
4. Adafruit Learning System. DRV8833 motor driver overview and pinouts.
<https://learn.adafruit.com/adafruit-drv8833-dc-stepper-motor-driver-breakout-board/overview> and
<https://learn.adafruit.com/adafruit-drv8833-dc-stepper-motor-driver-breakout-board/pinouts> . Guide published 2016, page checked 2026-09-11.
5. Texas Instruments. DRV8833 datasheet, SLVSAR1E, July 2015 revision.
<https://www.ti.com/lit/ds/symlink/drv8833.pdf> . Checked 2026-09-11.
6. Bioenno Power. BLF-1206A pack and compatible charger specifications.
<https://www.bioennopower.com/products/12v-6ah-lifepo4-battery-pvc> . Checked 2026-09-11.
7. Adafruit Industries. Lithium Ion Battery Pack, product 353. <https://www.adafruit.com/product/353> . Checked 2026-09-11.
8. Adafruit Industries. TowerPro MG92B servo, product 2307. <https://www.adafruit.com/product/2307> . Checked 2026-09-11.
9. Adafruit Industries. FeeTech FS5103R servo, product 154. <https://www.adafruit.com/product/154> . Checked 2026-09-11.
10. LilyGO. TTGO-T-Display original board repository, pinout, schematic and CAD.
<https://github.com/Xinyuan-LilyGO/TTGO-T-Display> . Checked 2026-09-11.

11. Espressif Systems. Arduino ESP32 Wi-Fi API.

<https://docs.espressif.com/projects/arduino-esp32/en/latest/api/wifi.html> . Current API concepts checked 2026-09-11; the compiled firmware pins the older compatible platform version in `firmware/platformio.ini`.

Original robot voice collection

The 24 MP3 files are in `../firmware/data/audio/`. They are included there so the same files can be uploaded to the robot's LittleFS flash filesystem. `catalog.json` records each exact transcript, duration, size and SHA-256. `phrases.csv` is the editable source list.

These are newly synthesized robot lines, not audio extracted from Doctor Who, a BBC recording, or a particular actor's performance. The text was written for this build. The source is the installed Microsoft David Desktop voice through Windows System.Speech. A low oscillator adds a metallic sound. The collection includes brief familiar words such as "Exterminate" alongside original control and character phrases. No downloaded sound samples are used.

The supplied format is mono MP3, 22,050 Hz, 32 kbit/s. Playback uses the ESP32 and the MAX98357A amplifier in the body. The speaker and all its wires stay stationary during head rotation. The controller lists files automatically. The files are manual sound buttons; the presence of a warning phrase does not mean firmware automatically plays that warning.

To recreate the collection on this Windows machine:

```
powershell -NoProfile -ExecutionPolicy Bypass -File C:/dev/robots/dalek/scripts/generate_voices.ps1
```

This requires the installed Microsoft desktop voice, Python with NumPy, and FFmpeg/FFprobe on PATH. It runs once. It registers no scheduled task. It creates a unique temporary WAV directory and removes only that directory when finished. The persistent outputs are the MP3s and catalog.

To replace a line, edit its row in `phrases.csv`, regenerate, rebuild the filesystem with PlatformIO, and upload that filesystem while actuator power is off. The waveform effect uses a 0.86 playback-rate multiplier, nonlinear amplitude compression, 30 Hz ring modulation, a quiet 45 ms echo, and a 170-3400 Hz frequency band. Peak amplitude is limited before MP3 encoding. Start with amplifier volume low and check the speaker for distortion.

To add your own recordings, use lowercase filenames with letters, numbers, underscores or hyphens and the `.mp3` suffix. Keep the entire flash filesystem inside its partition. Use only recordings you have permission to use. Character names and third-party voice/tool software retain their respective ownership; this package does not grant rights in third-party material.

Digital verification and physical limits

The software and fabrication files were checked on the design machine. These checks establish digital consistency, not operation of a completed robot. The first physical build must complete every gate in `assembly.md`.

Firmware and controls

Command:

```
pio run -d C:/dev/robots/dalek/firmware
```

Result: SUCCESS. RAM 45,416 / 327,680 bytes. Application flash 994,513 / 2,031,616 bytes. The build uses the original LilyGo T-Display board definition and Arduino-ESP32 2.0.14.

Command:

```
pio run -d C:/dev/robots/dalek/firmware -t buildfs
```

Result: SUCCESS. The filesystem image contains the local HTML/JavaScript interface and all 24 MP3s. It fits the 2,097,152-byte LittleFS partition. It has not been uploaded to a physical board during this run.

Commands:

```
node --check firmware/data/app.js
```

```
node firmware/test/test_ui.js
```

Results: JavaScript syntax passed. Browser test returned:

```
PASS: complete fresh commands, pointer cancel, blur, connection loss, late arm reply, no replay or auto-arm
```

Native test commands:

```
$env:PATH = 'C:/msys64/mingw64/bin;' + $env:PATH
```

```
g++ -std=c++11 -Wall -Wextra -Werror -I firmware/include firmware/test/test_control.cpp -o "$env:TEMP/dalek-control-test.exe"
```

```
& "$env:TEMP/dalek-control-test.exe"
```

Result:

```
PASS: parsing, boot lockout, lease replay, timeout, reconnect, fault interlock, rollover, ramp/reversal
```

The C++ test uses the same command-state and ramp code as the ESP32 firmware. The Node test executes the browser JavaScript with a simulated page and network. These tests do not measure Wi-Fi range, real browser timing on a phone, physical stopping distance, or servo response to loss of pulses.

Audio

Command:

```
powershell -NoProfile -ExecutionPolicy Bypass -File C:/dev/robots/dalek/scripts/generate_voices.ps1
```

Result:

```
PASS: 24 MP3s decoded; 407724 bytes; 98.7 seconds
```

Every file was encoded and decoded by FFmpeg. FFprobe checked MP3, mono and 22,050 Hz. Exact durations, file sizes and SHA-256 values are in `audio/catalog.json`. Speaker playback on the actual ESP32 remains a physical commissioning check.

Electrical consistency

Command:

```
python electronics/generate.py
```

Result: 168 point-to-point wiring rows, four parseable SVG circuit sheets, and calculated power/ADC values. Root and an independent firmware reviewer checked the actual circuit net names against firmware GPIOs, enable polarities, PCA9685 channel order, AHCT buffer pins, ADC ratio and thresholds, and audio wiring. The reviewed values agree.

The circuit sheets were rendered and visually inspected. An initial SVG rendering fault that hid CSS-styled wires was corrected in the generator using explicit SVG presentation attributes. The final diagrams contain visible connections. Electrical behavior, noise, supply sag, and actual servo current are still unmeasured.

CAD and printing

Use the final machine-readable report at `cad/validation.json` and print list at `bom/printed-parts.csv`. The exact check command is:

```
python scripts/export_cad.py --check-only
```

The exporter checks closed surfaces, consistent mesh winding, positive volume, connected physical solids and the conservative 300 x 300 x 300 mm print envelope. Closed internal cavities are distinguished from loose external solids. The final report records the actual results and solid-material mass bound for every part. Original CAD and every final STL are included.

Final check result: `mesh_count: 35, all_pass: true, printed_piece_count: 151, solid_material_upper_bound_g: 1988.1`. The complete listed print set includes a fit coupon and spare standoffs. Overall CAD height is 541 mm. The base diameter is 360 mm. Each STL is exported in pieces that fit the stated printer envelope. No printed weight or assembled weight is claimed as measured.

An actual Bambu Studio H2D slice of `fit_coupon.stl` also passed. The isolated one-time command, full profile names, Bambu Studio version 02.08.02.61 and exact STL SHA-256 are in `cad/h2d-slice-check.json`. The slicer returned `return_code: 0, error_string: Success`, and no warnings. It generated G-code at 0.20 mm layers, two walls and 15% infill. Predicted material was 12.1266 g and predicted total time was 1456.8 seconds. This was a PLA preflight slice; the functional fit coupon must still be printed in the material used for its matching structural parts. No printer was connected, and no actual print was made. The remaining parts were checked as meshes; they have not all been sliced during this run.

Independent coordinate reviews checked the chassis hole pattern, wheel and motor-mount clearance, plunger connectivity, rear board retention, arm attachment faces, and head bearing/hub stack. Identified geometric errors were corrected before the final export. These checks cannot replace test printing: the exact gearbox moulding, servo horn pattern, screen fit, layer shrinkage, bearing press fit and fastener access require physical checks.

What remains unverified

No physical robot was fabricated, wired, powered, flashed or driven during this run. Actual print duration, completed mass, fit, stability, motor current, servo current, noise, temperature, battery runtime, head friction, wireless range and stopping distance remain unverified. Availability of the specified wheels and head servo also requires procurement. The manual provides explicit staged checks for all of these items.

No cloud deployment was performed because the robot operates locally and this repository has no deployment workflow for the design package. Delivery consists of source and fabrication files committed and pushed to main, plus the reviewed design PDF sent by email.

Electrical bill of materials

bom\electronics.csv

- 1. reference:** M1-M4 | **quantity:** 4 | **item:** TT geared drive motor | **manufacturer part:** Adafruit 3777 | **unit usd:** 2.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/3777> | **notes:** Required motor; 3-6V; each uses its own 1A limited bridge
- 2. reference:** W1-W4 | **quantity:** 4 | **item:** Orange clear TT wheel | **manufacturer part:** Adafruit 3766 | **unit usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** Out of stock | **url:** <https://www.adafruit.com/product/3766> | **notes:** Required wheel; 63x29mm; keep exact part; obtain stock before wheel-fit signoff
- 3. reference:** U1 | **quantity:** 1 | **item:** Original TTGO T-Display ESP32 1.14 inch | **manufacturer part:** LILYGO TTGO T-Display | **unit usd:** 25.00 | **price basis:** allowance | **availability checked 2026 09 11:** Exact revision and stock must be checked | **url:** <https://github.com/Xinyuan-LilyGO/TTGO-T-Display> | **notes:** Existing user board can remove this cost; not T-Display-S3 or AMOLED
- 4. reference:** U5-U6 | **quantity:** 2 | **item:** DRV8833 dual motor driver | **manufacturer part:** Adafruit 3297 | **unit usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/3297> | **notes:** Retain two stock 0.2ohm sense resistors on each board
- 5. reference:** U10 | **quantity:** 1 | **item:** PCA9685 servo PWM board | **manufacturer part:** Adafruit 815 | **unit usd:** 14.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/815> | **notes:** Remove OE pull-down; VCC3.3V; leave V+ unconnected
- 6. reference:** SV0-SV3 | **quantity:** 4 | **item:** MG92B positional metal gear micro servo | **manufacturer part:** Adafruit 2307 | **unit usd:** 11.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/2307> | **notes:** Includes horns and horn screws; needs 5V control signals
- 7. reference:** SV4 | **quantity:** 1 | **item:** FS5103R continuous rotation head servo | **manufacturer part:** Adafruit 154 | **unit usd:** 11.95 | **price basis:** listed | **availability checked 2026 09 11:** Out of stock | **url:** <https://www.adafruit.com/product/154> | **notes:** Includes horns; speed control only; no angle feedback
- 8. reference:** U8-U9 | **quantity:** 2 | **item:** 74AHCT125 DIP14 level buffer | **manufacturer part:** Adafruit 1787 | **unit usd:** 1.50 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1787> | **notes:** Use AHCT with powered-off-input tolerance; verify actual IC marking against datasheet
- 9. reference:** U2-U3 | **quantity:** 2 | **item:** 5V5.5A buck regulator | **manufacturer part:** Pololu D36V50F5 4091 | **unit usd:** 39.95 | **price basis:** listed | **availability checked 2026 09 11:** Rationed active and preferred; backorders allowed | **url:** <https://www.pololu.com/product/4091> | **notes:** Separate motor and servo regulators; 25.4x25.4x9.5mm; no outputs paralleled
- 10. reference:** U4 | **quantity:** 1 | **item:** 5V1.2A logic buck regulator | **manufacturer part:** Adafruit MPM3610 4739 | **unit usd:** 6.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/4739> | **notes:** Input max21V; solder power pins; EN left open
- 11. reference:** B1 | **quantity:** 1 | **item:** Protected12V6Ah72WhLiFePO4 battery | **manufacturer part:** Bioenno BLF-1206A | **unit usd:** 79.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/12v-6ah-lifepo4-battery-pvc> | **notes:** 12A continuous; 0.7kg; PP30 discharge; padded pocket 116x71x76mm minimum
- 12. reference:** CH1 | **quantity:** 1 | **item:** Matched14.6V2A LiFePO4 charger | **manufacturer part:** Bioenno BPC-1502DC | **unit usd:** 21.99 | **price basis:** listed | **availability checked 2026 09 11:** Add to cart offered | **url:** <https://www.bioennopower.com/products/lithium-12v-2a-amp-lifepo4-battery-charger> | **notes:** Use factory battery charge connector; unplug and remove battery to charge
- 13. reference:** U7 | **quantity:** 1 | **item:** MAX98357A mono I2S amplifier | **manufacturer part:** Adafruit 3006 | **unit usd:** 5.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/3006> | **notes:** GAIN100k to 5V gives 3dB; fixed digital gain 0.28
- 14. reference:** SPK1 | **quantity:** 1 | **item:** 8ohm1W3inch speaker | **manufacturer part:** Adafruit 1313 | **unit usd:** 1.95 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1313> | **notes:** 77.8x77.8x25.49mm; never ground either speaker lead
- 15. reference:** S1-S2 | **quantity:** 2 | **item:** Panel SPST ON-OFF disconnect | **manufacturer part:** NKK S1A | **unit usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:** https://www.nkkswitches.com/wp-content/themes/impress-blank/search/inc/part.php?part_no=S1A | **notes:** 20A30VDC; 12.5mm hole;

includes M12 panel hardware; OFF label required

- 16. reference:** FH1-FH4 | **quantity:** 4 | **item:** Inline ATO fuse holder16AWG | **manufacturer part:** Littelfuse 0FHA0001ZXJ | **unit**
usd: 4.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:**
<https://info.littelfuse.com/ato-fha-series-landing-page> | **notes:** 20A holder; insulate leads and restrain wiring
- 17. reference:** F1 | **quantity:** 1 | **item:** ATO7.5A32V fuse | **manufacturer part:** Littelfuse 025707.5 | **unit usd:** 1.00 | **price basis:**
allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:**
<https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Main pack fuse
within100mm of connector
- 18. reference:** F2-F3 | **quantity:** 2 | **item:** ATO3A32V fuse | **manufacturer part:** Littelfuse 0257003 | **unit usd:** 1.00 | **price basis:**
allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:**
<https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input branch fuses for
motor and servo converters
- 19. reference:** F4 | **quantity:** 1 | **item:** ATO1A32V fuse | **manufacturer part:** Littelfuse 0257001 | **unit usd:** 1.00 | **price basis:**
allowance | **availability checked 2026 09 11:** Distributor quote needed | **url:**
<https://www.littelfuse.com/products/fuses-overcurrent-protection/fuses/automotive-passenger-car/blade-fuses> | **notes:** Input fuse for logic
converter
- 20. reference:** PP30 | **quantity:** 1 | **item:** Anderson Powerpole mating red-black pair with contacts | **manufacturer part:** PP30 genuine
 housings and30A contacts | **unit usd:** 6.00 | **price basis:** allowance | **availability checked 2026 09 11:** Distributor quote needed |
url: <https://www.andersonpower.com/us/en/resources/PowerpoleResourcesPage.html> | **notes:** Use contacts matched to18AWG; do not cut
factory battery lead
- 21. reference:** PROTO | **quantity:** 2 | **item:** Half-size solderable prototype board | **manufacturer part:** Adafruit 1609 | **unit usd:** 4.50
 | **price basis:** listed | **availability checked 2026 09 11:** In stock | **url:** <https://www.adafruit.com/product/1609> | **notes:** One buffer
board; one low-current divider board; power distribution uses separate heavy wire
- 22. reference:** SOCKET | **quantity:** 2 | **item:** DIP14 IC socket | **manufacturer part:** 14pin0.3inch socket | **unit usd:** 0.75 | **price**
basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Protects buffers during soldering
- 23. reference:** R1-R6 R12 R20-R24 R40-R44 | **quantity:** 17 | **item:** 10kohm1percent0.25W resistor | **manufacturer part:** Metal film
axial | **unit usd:** 0.10 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Pull-downs and
OE pull-up
- 24. reference:** R10 R14 | **quantity:** 2 | **item:** 100kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10
 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider top and amplifier gain
resistor
- 25. reference:** R11 | **quantity:** 1 | **item:** 22kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 |
price basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Battery divider bottom; never replace
with33k
- 26. reference:** R13 | **quantity:** 1 | **item:** 15kohm1percent0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 |
price basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator sense bottom resistor
- 27. reference:** R30-R34 | **quantity:** 5 | **item:** 220ohm0.25W resistor | **manufacturer part:** Metal film axial | **unit usd:** 0.10 | **price**
basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Servo signal series resistors
- 28. reference:** R15-R16 | **quantity:** 2 | **item:** 100ohm1W resistor | **manufacturer part:** Metal film flameproof axial | **unit usd:** 0.40 |
price basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Actuator rail discharge; dissipates0.25W
each during use
- 29. reference:** C8-C11 MOTOR_CAPS | **quantity:** 8 | **item:** 100nF25Vorhigher ceramic capacitor | **manufacturer part:** X7R radial |
unit usd: 0.15 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Four motor
suppression;two buffer bypass;two ADC filters
- 30. reference:** C1-C2 | **quantity:** 2 | **item:** 220uF25V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75
 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Motor and servo converter input;
polarity critical
- 31. reference:** C3 | **quantity:** 1 | **item:** 100uF25V capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.50 | **price**
basis: allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Logic converter input

- 32. reference:** C4-C5 C7 | **quantity:** 3 | **item:** 470uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 0.75 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** One at each driver and one at logic output
- 33. reference:** C6 | **quantity:** 1 | **item:** 1000uF10V lowESR capacitor | **manufacturer part:** 105C radial electrolytic | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Servo distribution block; polarity critical
- 34. reference:** POWER_WIRE | **quantity:** 4 | **item:** 18AWG flexible stranded copper wire metre | **manufacturer part:** Silicone insulation red-black-orange | **unit usd:** 1.50 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Combined total4m; trim to actual path plus service loop
- 35. reference:** BRANCH_WIRE | **quantity:** 5 | **item:** 22AWG flexible stranded wire metre | **manufacturer part:** Red and black copper | **unit usd:** 1.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Individual servo and logic branches
- 36. reference:** SIGNAL_WIRE | **quantity:** 5 | **item:** 26AWG stranded signal wire metre | **manufacturer part:** Assorted color copper | **unit usd:** 0.60 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Keep I2C and I2S runs short under200mm where possible
- 37. reference:** CONNECTORS | **quantity:** 1 | **item:** Connector and termination allowance | **manufacturer part:** Insulated terminal blocks and ferrules and servo header extensions | **unit usd:** 18.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Power blocks rated>=10A30VDC;five servo3pin sockets;removable run-power link;four2pin motor plugs
- 38. reference:** INSULATION | **quantity:** 1 | **item:** Electrical assembly consumables | **manufacturer part:** Heatshrink cable ties labels solder and foam | **unit usd:** 12.00 | **price basis:** allowance | **availability checked 2026 09 11:** Generic stock not checked | **notes:** Insulation and strain relief are required

Mechanical hardware bill of materials

bom\hardware.csv

- 1. category:** fastener | **item:** M3x8 socket screw | **quantity:** 48 | **unit:** piece | **location:** Hemispheres | **specification:** Metric coarse thread; install from inside shell
- 2. category:** fastener | **item:** M3x10 socket screw | **quantity:** 4 | **unit:** piece | **location:** Dome to head plate | **specification:** Metric coarse thread; captive nuts in dome lugs
- 3. category:** fastener | **item:** M3x12 socket screw | **quantity:** 8 | **unit:** piece | **location:** Bumper seams | **specification:** Two per seam
- 4. category:** fastener | **item:** M3x12 socket screw | **quantity:** 16 | **unit:** piece | **location:** Skirt vertical seams | **specification:** Two per seam per tier
- 5. category:** fastener | **item:** M3x12 socket screw | **quantity:** 24 | **unit:** piece | **location:** Skirt horizontal joins | **specification:** 12 lower-to-upper plus 12 upper-to-shoulder
- 6. category:** fastener | **item:** M3x12 socket screw | **quantity:** 4 | **unit:** piece | **location:** Shoulder side seams | **specification:** Two per side
- 7. category:** fastener | **item:** M3x12 socket screw | **quantity:** 6 | **unit:** piece | **location:** Speaker | **specification:** Two foot bolts plus four speaker-tab bolts
- 8. category:** fastener | **item:** M3x12 socket screw | **quantity:** 3 | **unit:** piece | **location:** Neck adapter to shoulder | **specification:** Radius 110 at 30/150/270 degrees
- 9. category:** fastener | **item:** M3x12 socket screw | **quantity:** 4 | **unit:** piece | **location:** Bearing tower to neck deck | **specification:** Radius 24 at 45/135/225/315 degrees
- 10. category:** fastener | **item:** M3x12 socket screw | **quantity:** 4 | **unit:** piece | **location:** Head servo cradle to neck deck | **specification:** Slotted belt-tension mount
- 11. category:** fastener | **item:** M3x16 socket screw | **quantity:** 24 | **unit:** piece | **location:** Chassis splice plates | **specification:** Two per plate
- 12. category:** fastener | **item:** M3x16 socket screw | **quantity:** 12 | **unit:** piece | **location:** Bumper/chassis/lower-skirt sandwich | **specification:** Three per quadrant
- 13. category:** fastener | **item:** M3x16 socket screw | **quantity:** 4 | **unit:** piece | **location:** Battery tray | **specification:** Heads above tray; nuts below chassis
- 14. category:** fastener | **item:** M3x16 socket screw | **quantity:** 8 | **unit:** piece | **location:** Arm bases | **specification:** Four per arm
- 15. category:** fastener | **item:** M3x16 socket screw | **quantity:** 4 | **unit:** piece | **location:** Head hub to head plate | **specification:** Four at 17mm radius
- 16. category:** fastener | **item:** M3x16 socket screw | **quantity:** 2 | **unit:** piece | **location:** Eye stalk | **specification:** Two at 9mm offset
- 17. category:** fastener | **item:** M3x16 socket screw | **quantity:** 2 | **unit:** piece | **location:** Decorative lamps | **specification:** One per lamp; nut captive in lamp base
- 18. category:** fastener | **item:** M3x25 socket screw | **quantity:** 4 | **unit:** piece | **location:** Rear display cassette | **specification:** Through frame feet shell and rear-cover posts
- 19. category:** fastener | **item:** M3x45 socket screw | **quantity:** 3 | **unit:** piece | **location:** Neck ring stack | **specification:** Through adapter rings six posts and neck deck
- 20. category:** fastener | **item:** M3x65 socket screw | **quantity:** 16 | **unit:** piece | **location:** Motor cradles | **specification:** Through 55mm pillars plus 5mm chassis
- 21. category:** fastener | **item:** M3x110 socket screw | **quantity:** 4 | **unit:** piece | **location:** Electronics deck | **specification:** Through chassis 95mm post and 3mm deck

- 22. category:** fastener | **item:** M3x8 plastic thread-forming screw | **quantity:** 3 | **unit:** piece | **location:** Bearing cap | **specification:** 2.6mm pilot holes; do not confuse with machine screws
- 23. category:** fastener | **item:** M3 hex nut | **quantity:** 220 | **unit:** piece | **location:** All M3 joints | **specification:** Includes approximately 16 spares; standard 5.5mm across flats
- 24. category:** fastener | **item:** M3 plain washer | **quantity:** 400 | **unit:** piece | **location:** Thin shell and structural joints | **specification:** Small outside diameter 6-7mm; use under head and nut on thin shell
- 25. category:** fastener | **item:** M2x10 machine screw | **quantity:** 10 | **unit:** piece | **location:** Supplied servo horns | **specification:** Two per horn for five servos; do not print splines
- 26. category:** fastener | **item:** M2 nut and washer set | **quantity:** 10 | **unit:** set | **location:** Supplied servo horns | **specification:** Metric coarse thread
- 27. category:** fastener | **item:** M2.5x16 machine screw | **quantity:** 32 | **unit:** piece | **location:** Electronics boards | **specification:** Through PCB 6mm spacer and deck; check each actual board hole
- 28. category:** fastener | **item:** M2.5 nut and washer set | **quantity:** 32 | **unit:** set | **location:** Electronics boards | **specification:** Nylon preferred where board clearances are small
- 29. category:** fastener | **item:** M8x70 hex-head bolt | **quantity:** 1 | **unit:** piece | **location:** Rotating head spindle | **specification:** Fully threaded or threaded end long enough for upper nut
- 30. category:** fastener | **item:** M8 nyloc nut | **quantity:** 1 | **unit:** piece | **location:** Head spindle | **specification:** Set gently against inner-race stack; do not clamp bearing outer races
- 31. category:** fastener | **item:** M8 standard washer | **quantity:** 1 | **unit:** piece | **location:** Top of head plate | **specification:** Only above printed head plate; never against bearing inner race
- 32. category:** fastener | **item:** 8x12x1mm precision shim washer | **quantity:** 2 | **unit:** piece | **location:** Bearing inner races | **specification:** 8mm ID 12mm OD 1mm thick; narrow OD must clear 16mm cap bore
- 33. category:** bearing | **item:** 608-2RS bearing | **quantity:** 2 | **unit:** piece | **location:** Head bearing tower | **specification:** 22mm OD 8mm ID 7mm width; ordinary skateboard bearing size
- 34. category:** belt | **item:** NBR round O-ring | **quantity:** 1 | **unit:** piece | **location:** Head belt | **specification:** 83mm ID 3mm cord; adjustable 72.5mm nominal shaft centre spacing
- 35. category:** retainer | **item:** 20mm hook-and-loop strap 300mm long | **quantity:** 2 | **unit:** piece | **location:** Battery | **specification:** Pass through tray slots; independent straps
- 36. category:** retainer | **item:** 2.5mm nylon cable tie 150-200mm long | **quantity:** 50 | **unit:** piece | **location:** Motors servos and cable routing | **specification:** At least 8 motor ties 8 arm-servo ties 2 head-servo ties plus routing and spares
- 37. category:** padding | **item:** Closed-cell EPDM foam sheet | **quantity:** 1 | **unit:** sheet | **location:** Battery servo and screen padding | **specification:** Approximately 200x150mm; 1mm 2mm 3mm thickness assortment; electrically nonconductive
- 38. category:** consumable | **item:** Thread-locking compound removable | **quantity:** 1 | **unit:** small bottle | **location:** Metal-to-metal joints only | **specification:** Keep away from printed plastic and bearings
- 39. category:** consumable | **item:** PLA filament | **quantity:** 1 | **unit:** kg | **location:** Cosmetic parts | **specification:** Lightweight bronze or painted; two 0.4mm shell walls
- 40. category:** consumable | **item:** PETG filament | **quantity:** 1.5 | **unit:** kg | **location:** Structural parts | **specification:** Includes support and fit-test allowance; actual usage comes from slicer
- 41. category:** tool | **item:** Metric hex keys and nut drivers | **quantity:** 1 | **unit:** set | **location:** Assembly | **specification:** M2/M2.5/M3/M8 compatible
- 42. category:** tool | **item:** Digital caliper | **quantity:** 1 | **unit:** piece | **location:** Fit checks | **specification:** Measure actual motors servo bodies board and bearing pockets before bulk printing
- 43. category:** tool | **item:** Drill bits 2.1mm 2.5mm 3.4mm | **quantity:** 1 | **unit:** set | **location:** Horn and clearance preparation | **specification:** Hand turn printed-hole clearing where possible; secure parts before drilling

Printed part quantities and orientations

bom\printed-parts.csv

1. part: fit_coupon | **quantity:** 1 | **material:** PETG | **x mm:** 90.0 | **y mm:** 50.0 | **z mm:** 5.0 | **volume mm3:** 19541.89 | **solid mass g:** 24.82 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Print first; gauge M3/M8/bearing/switch fits

2. part: chassis_quarter | **quantity:** 4 | **material:** PETG | **x mm:** 180.0 | **y mm:** 180.0 | **z mm:** 5.0 | **volume mm3:** 71993.63 | **solid mass g:** 91.43 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** 5mm deck; flip flat plates to mirror X/Y holes; do not rotate 90 degrees

3. part: bumper_quarter | **quantity:** 4 | **material:** PLA | **x mm:** 180.0 | **y mm:** 180.0 | **z mm:** 57.0 | **volume mm3:** 23335.26 | **solid mass g:** 28.94 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Lower edge down | **notes:** 15mm floor clearance

4. part: skirt_lower_quarter | **quantity:** 4 | **material:** PLA | **x mm:** 180.0 | **y mm:** 180.0 | **z mm:** 90.0 | **volume mm3:** 36542.72 | **solid mass g:** 45.31 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Lower flange down | **notes:** 90mm height; M3 seam joints

5. part: skirt_upper_quarter | **quantity:** 4 | **material:** PLA | **x mm:** 145.0 | **y mm:** 145.0 | **z mm:** 90.0 | **volume mm3:** 30596.17 | **solid mass g:** 37.94 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Lower flange down | **notes:** 90mm height; rear quarters removable

6. part: splice | **quantity:** 12 | **material:** PETG | **x mm:** 24.0 | **y mm:** 24.0 | **z mm:** 3.0 | **volume mm3:** 1674.14 | **solid mass g:** 2.13 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Three plates per chassis radial seam

7. part: motor_cradle | **quantity:** 4 | **material:** PETG | **x mm:** 51.0 | **y mm:** 76.0 | **z mm:** 55.0 | **volume mm3:** 21497.75 | **solid mass g:** 27.3 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Base down | **notes:** 55mm pillars; M3x65 through bolts

8. part: battery_tray | **quantity:** 1 | **material:** PETG | **x mm:** 124.0 | **y mm:** 80.0 | **z mm:** 15.0 | **volume mm3:** 45292.29 | **solid mass g:** 57.52 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** 116x72mm pocket; 2x20mm straps

9. part: deck_post | **quantity:** 4 | **material:** PETG | **x mm:** 9.0 | **y mm:** 9.0 | **z mm:** 95.0 | **volume mm3:** 5152.17 | **solid mass g:** 6.54 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Upright | **notes:** 95mm; M3x110 through bolt

10. part: electronics_deck | **quantity:** 1 | **material:** PETG | **x mm:** 180.0 | **y mm:** 120.0 | **z mm:** 3.0 | **volume mm3:** 47230.42 | **solid mass g:** 59.98 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** M2.5 insulated standoffs on slots

11. part: pcb_standoff | **quantity:** 32 | **material:** PETG | **x mm:** 7.0 | **y mm:** 7.0 | **z mm:** 6.0 | **volume mm3:** 191.75 | **solid mass g:** 0.24 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** 6mm insulated PCB spacers; quantity includes spares

12. part: shoulder_front | **quantity:** 1 | **material:** PLA | **x mm:** 240.0 | **y mm:** 120.0 | **z mm:** 110.0 | **volume mm3:** 82105.71 | **solid mass g:** 101.81 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Bottom flange down | **notes:** Flat front retains both arm brackets

13. part: shoulder_rear | **quantity:** 1 | **material:** PLA | **x mm:** 240.0 | **y mm:** 120.0 | **z mm:** 110.0 | **volume mm3:** 56154.98 | **solid mass g:** 69.63 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Bottom flange down | **notes:** Display and two switches; removable

14. part: screen_frame | **quantity:** 1 | **material:** PETG | **x mm:** 88.0 | **y mm:** 53.0 | **z mm:** 15.0 | **volume mm3:** 23706.26 | **solid mass g:** 30.11 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** As exported | **notes:** 52.2x26 board pocket; USB side opening

15. part: screen_clamp | **quantity:** 1 | **material:** PETG | **x mm:** 88.0 | **y mm:** 53.0 | **z mm:** 8.0 | **volume mm3:** 13279.19 | **solid mass g:** 16.86 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Rear cover; foam pads contact PCB edges

16. part: arm_base | **quantity:** 2 | **material:** PETG | **x mm:** 44.0 | **y mm:** 46.0 | **z mm:** 48.0 | **volume mm3:** 21680.39 | **solid mass g:** 27.53 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Base down | **notes:** MG92B upright cradle; two cable ties

17. part: pitch_carrier | **quantity:** 2 | **material:** PETG | **x mm:** 49.0 | **y mm:** 48.0 | **z mm:** 44.0 | **volume mm3:** 13977.47 | **solid mass g:** 17.75 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Base down | **notes:** MG92B axis horizontal; supplied yaw horn

18. part: plunger_arm | **quantity:** 1 | **material:** PLA | **x mm:** 142.0 | **y mm:** 42.0 | **z mm:** 42.0 | **volume mm3:** 18122.12 | **solid mass g:** 22.47 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 1 | **pass check:** True | **orientation:** As exported; supports under cup | **notes:** Decorative hollow tube; no load

19. part: gun_arm | **quantity:** 1 | **material:** PLA | **x mm:** 135.0 | **y mm:** 28.0 | **z mm:** 19.0 | **volume mm3:** 17608.87 | **solid mass g:** 21.84 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 1 | **pass check:** True | **orientation:** As exported; supports under rods | **notes:** Solid decorative muzzle; no projectile function

20. part: neck_adapter | **quantity:** 1 | **material:** PLA | **x mm:** 240.0 | **y mm:** 240.0 | **z mm:** 4.0 | **volume mm3:** 14652.34 | **solid mass g:** 18.17 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Shoulder 110mm PCD-radius to neck104mm radius

21. part: neck_ring | **quantity:** 2 | **material:** PLA | **x mm:** 216.0 | **y mm:** 216.0 | **z mm:** 4.0 | **volume mm3:** 18247.38 | **solid mass g:** 22.63 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Three supports at30/150/270 degrees

22. part: neck_post | **quantity:** 6 | **material:** PETG | **x mm:** 9.0 | **y mm:** 9.0 | **z mm:** 11.0 | **volume mm3:** 593.09 | **solid mass g:** 0.75 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Upright | **notes:** 11mm spacing

23. part: neck_deck | **quantity:** 1 | **material:** PETG | **x mm:** 216.0 | **y mm:** 216.0 | **z mm:** 4.0 | **volume mm3:** 48922.9 | **solid mass g:** 62.13 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Bearing and belt motor support

24. part: bearing_tower | **quantity:** 1 | **material:** PETG | **x mm:** 58.0 | **y mm:** 58.0 | **z mm:** 26.0 | **volume mm3:** 21792.14 | **solid mass g:** 27.68 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flange down | **notes:** 608 bearings22x8x7; lower bearing inserts from bottom

25. part: bearing_cap | **quantity:** 1 | **material:** PETG | **x mm:** 34.0 | **y mm:** 34.0 | **z mm:** 3.0 | **volume mm3:** 2042.29 | **solid mass g:** 2.59 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** 3xM3x8 thread-forming screws

26. part: inner_spacer | **quantity:** 1 | **material:** PETG | **x mm:** 11.5 | **y mm:** 11.5 | **z mm:** 12.0 | **volume mm3:** 586.98 | **solid mass g:** 0.75 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** 12mm between bearing inner races

27. part: head_hub | **quantity:** 1 | **material:** PETG | **x mm:** 64.0 | **y mm:** 64.0 | **z mm:** 25.0 | **volume mm3:** 30423.06 | **solid mass g:** 38.64 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Foot down; supports under first rim | **notes:** M8 spindle;3mm round belt groove

28. part: head_servo_mount | **quantity:** 1 | **material:** PETG | **x mm:** 36.0 | **y mm:** 72.0 | **z mm:** 21.0 | **volume mm3:** 15427.38 | **solid mass g:** 19.59 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** FS5103R clamp; belt-tension slots

29. part: servo_pulley | **quantity:** 1 | **material:** PETG | **x mm:** 25.0 | **y mm:** 25.0 | **z mm:** 9.0 | **volume mm3:** 2998.48 | **solid mass g:** 3.81 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Attach supplied horn; no printed spline

30. part: head_plate | **quantity:** 1 | **material:** PETG | **x mm:** 216.0 | **y mm:** 216.0 | **z mm:** 3.0 | **volume mm3:** 23528.5 | **solid mass g:** 29.88 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Spokes retain dome with4M3 bolts

31. part: dome | **quantity:** 1 | **material:** PLA | **x mm:** 220.0 | **y mm:** 220.0 | **z mm:** 64.922 | **volume mm3:** 70343.13 | **solid mass g:** 87.23 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Open base down; tree support inside | **notes:** 220mm diameter hollow dome; stationary wiring excluded

32. part: eye | **quantity:** 1 | **material:** PLA | **x mm:** 32.0 | **y mm:** 32.0 | **z mm:** 92.0 | **volume mm3:** 18107.54 | **solid mass g:** 22.45 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 1 | **pass check:** True | **orientation:** Base flange down | **notes:** Rear two M3 bolts; black pupil paint

33. part: lamp | **quantity:** 2 | **material:** Translucent PLA | **x mm:** 25.0 | **y mm:** 25.0 | **z mm:** 22.0 | **volume mm3:** 5605.06 | **solid mass g:** 6.95 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Flat | **notes:** Decorative unlit lamps; no head wires

34. part: hemisphere | **quantity:** 48 | **material:** PLA | **x mm:** 25.969 | **y mm:** 25.969 | **z mm:** 12.881 | **volume mm3:** 1703.99 | **solid mass g:** 2.11 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** Open circular face down | **notes:** M3 captive nut; print6first to confirm seam access

35. part: speaker_mount | **quantity:** 1 | **material:** PETG | **x mm:** 90.0 | **y mm:** 34.0 | **z mm:** 90.0 | **volume mm3:** 24854.6 | **solid mass g:** 31.57 | **watertight:** True | **connected components:** 1 | **enclosed void surfaces:** 0 | **pass check:** True | **orientation:** As exported | **notes:** 85mm speaker envelope; bolts to deck front

Point-to-point wiring schedule

electronics\wiring.csv

1. **net:** BAT_POS | **origin:** B1.PP30+ | **destination:** F1.IN | **wire awg:** 18 | **color:** red | **note:** 7.5A fuse within 100mm of pack
2. **net:** BAT_FUSED | **origin:** F1.OUT | **destination:** S1.1 | **wire awg:** 18 | **color:** red
3. **net:** PACK_SW | **origin:** S1.3 | **destination:** S2.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
4. **net:** PACK_SW | **origin:** S1.3 | **destination:** F4.IN | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
5. **net:** PACK_SW | **origin:** S1.3 | **destination:** R10.1 | **wire awg:** 18 | **color:** red | **note:** NKK S1A ON connects terminals1 and3
6. **net:** ACT_IN | **origin:** S2.3 | **destination:** F2.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
7. **net:** ACT_IN | **origin:** S2.3 | **destination:** F3.IN | **wire awg:** 18 | **color:** orange | **note:** Physical actuator power disconnect; logic stays powered
8. **net:** MOTOR_IN | **origin:** F2.OUT | **destination:** U2.VIN | **wire awg:** 18 | **color:** orange | **note:** F2 3A
9. **net:** SERVO_IN | **origin:** F3.OUT | **destination:** U3.VIN | **wire awg:** 18 | **color:** orange | **note:** F3 3A
10. **net:** LOGIC_IN | **origin:** F4.OUT | **destination:** U4.IN | **wire awg:** 22 | **color:** red | **note:** F4 1A
11. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U5.VMotor | **wire awg:** 18 | **color:** orange
12. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** U6.VMotor | **wire awg:** 18 | **color:** orange
13. **net:** 5V_MOTOR | **origin:** U2.VOUT | **destination:** R12.1 | **wire awg:** 18 | **color:** orange
14. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** J_SERVO.2 | **wire awg:** 18 | **color:** red | **note:** Five separate servo branches; never feed through PCA9685 V+ traces
15. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U8.14 | **wire awg:** 18 | **color:** red | **note:** Five separate servo branches; never feed through PCA9685 V+ traces
16. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U9.14 | **wire awg:** 18 | **color:** red | **note:** Five separate servo branches; never feed through PCA9685 V+ traces
17. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** J_USB_ISO.1 | **wire awg:** 22 | **color:** red
18. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** U7.VIN | **wire awg:** 22 | **color:** red
19. **net:** 5V_LOGIC | **origin:** U4.OUT | **destination:** R14.1 | **wire awg:** 22 | **color:** red
20. **net:** ESP_5V | **origin:** J_USB_ISO.2 | **destination:** U1.5V | **wire awg:** 22 | **color:** red | **note:** Removable run-power link; OPEN and unplug battery before USB
21. **net:** 3V3 | **origin:** U1.3V3 | **destination:** U10.VCC | **wire awg:** 26 | **color:** red
22. **net:** 3V3 | **origin:** U1.3V3 | **destination:** R6.1 | **wire awg:** 26 | **color:** red
23. **net:** GND | **origin:** B1.PP30- | **destination:** U2.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
24. **net:** GND | **origin:** B1.PP30- | **destination:** U3.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
25. **net:** GND | **origin:** B1.PP30- | **destination:** U4.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
26. **net:** GND | **origin:** B1.PP30- | **destination:** U5.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board
27. **net:** GND | **origin:** B1.PP30- | **destination:** U6.GND | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

28. **net:** GND | **origin:** B1.PP30- | **destination:** J_SERVO.1 | **wire awg:** 18 | **color:** black | **note:** Star ground; power returns do not pass through ESP32 or PCA board

29. **net:** GND | **origin:** GROUND_STAR | **destination:** U1.GND | **wire awg:** 22 | **color:** black

30. **net:** GND | **origin:** GROUND_STAR | **destination:** U7.GND | **wire awg:** 22 | **color:** black

31. **net:** GND | **origin:** GROUND_STAR | **destination:** U8.7 | **wire awg:** 22 | **color:** black

32. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.7 | **wire awg:** 22 | **color:** black

33. **net:** GND | **origin:** GROUND_STAR | **destination:** U10.GND | **wire awg:** 22 | **color:** black

34. **net:** GND | **origin:** B1.PP30- | **destination:** GROUND_STAR | **wire awg:** 18 | **color:** black

35. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.AIN1 | **wire awg:** 26 | **color:** signal

36. **net:** LEFT_IN1 | **origin:** U1.GPIO25 | **destination:** U5.BIN1 | **wire awg:** 26 | **color:** signal

37. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.AIN2 | **wire awg:** 26 | **color:** signal

38. **net:** LEFT_IN2 | **origin:** U1.GPIO26 | **destination:** U5.BIN2 | **wire awg:** 26 | **color:** signal

39. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.AIN1 | **wire awg:** 26 | **color:** signal

40. **net:** RIGHT_IN1 | **origin:** U1.GPIO32 | **destination:** U6.BIN1 | **wire awg:** 26 | **color:** signal

41. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.AIN2 | **wire awg:** 26 | **color:** signal

42. **net:** RIGHT_IN2 | **origin:** U1.GPIO33 | **destination:** U6.BIN2 | **wire awg:** 26 | **color:** signal

43. **net:** LEFT_IN1 | **origin:** LEFT_IN1 | **destination:** R1.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

44. **net:** GND | **origin:** GND | **destination:** R1.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

45. **net:** LEFT_IN2 | **origin:** LEFT_IN2 | **destination:** R2.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

46. **net:** GND | **origin:** GND | **destination:** R2.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

47. **net:** RIGHT_IN1 | **origin:** RIGHT_IN1 | **destination:** R3.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

48. **net:** GND | **origin:** GND | **destination:** R3.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

49. **net:** RIGHT_IN2 | **origin:** RIGHT_IN2 | **destination:** R4.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

50. **net:** GND | **origin:** GND | **destination:** R4.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

51. **net:** MOTOR_ENABLE | **origin:** MOTOR_ENABLE | **destination:** R5.1 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

52. **net:** GND | **origin:** GND | **destination:** R5.2 | **wire awg:** component | **note:** 10k 1% 0.25W pull-down

53. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U5.SLP | **wire awg:** 26 | **color:** signal

54. **net:** MOTOR_ENABLE | **origin:** U1.GPIO12 | **destination:** U6.SLP | **wire awg:** 26 | **color:** signal

55. **net:** SDA | **origin:** U1.GPIO21 | **destination:** U10.SDA | **wire awg:** 26 | **color:** signal

56. **net:** SCL | **origin:** U1.GPIO22 | **destination:** U10.SCL | **wire awg:** 26 | **color:** signal

57. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U10.OE | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

58. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** R6.2 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

59. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.1 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

60. **net:** SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.4 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

61. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.10 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

62. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U8.13 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

63. net: SERVO_DISABLE | **origin:** U1.GPIO27 | **destination:** U9.1 | **wire awg:** 26 | **color:** signal | **note:** R6 10k to3V3; REMOVE PCA board OE pull-down

64. net: I2S_BCLK | **origin:** U1.GPIO17 | **destination:** U7.BCLK | **wire awg:** 26 | **color:** signal

65. net: I2S_LRC | **origin:** U1.GPIO13 | **destination:** U7.LRC | **wire awg:** 26 | **color:** signal

66. net: I2S_DATA | **origin:** U1.GPIO15 | **destination:** U7.DIN | **wire awg:** 26 | **color:** signal | **note:** GPIO15 strap: amplifier INPUT only; no pull-up/down added

67. net: BAT_ADC | **origin:** R10.2 | **destination:** R11.1 | **wire awg:** 26 | **color:** signal

68. net: BAT_ADC | **origin:** R10.2 | **destination:** U1.GPIO39 | **wire awg:** 26 | **color:** signal

69. net: BAT_ADC | **origin:** R10.2 | **destination:** C10.1 | **wire awg:** 26 | **color:** signal

70. net: GND | **origin:** GROUND_STAR | **destination:** R11.2 | **wire awg:** 26 | **color:** signal

71. net: GND | **origin:** GROUND_STAR | **destination:** C10.2 | **wire awg:** 26 | **color:** signal

72. net: ACT_OK | **origin:** R12.2 | **destination:** R13.1 | **wire awg:** 26 | **color:** signal

73. net: ACT_OK | **origin:** R12.2 | **destination:** U1.GPIO36 | **wire awg:** 26 | **color:** signal

74. net: ACT_OK | **origin:** R12.2 | **destination:** C11.1 | **wire awg:** 26 | **color:** signal

75. net: GND | **origin:** GROUND_STAR | **destination:** R13.2 | **wire awg:** 26 | **color:** signal

76. net: GND | **origin:** GROUND_STAR | **destination:** C11.2 | **wire awg:** 26 | **color:** signal

77. net: AMP_GAIN | **origin:** R14.2 | **destination:** U7.GAIN | **wire awg:** 26 | **color:** signal | **note:** R14 100k to5V_LOGIC sets3dB

78. net: SPK_P | **origin:** U7.OUT+ | **destination:** SPK1.+ | **wire awg:** 24 | **color:** yellow | **note:** Floating bridge output; NEVER ground speaker leads

79. net: SPK_N | **origin:** U7.OUT- | **destination:** SPK1.- | **wire awg:** 24 | **color:** blue

80. net: LEFT_FRONT_+ | **origin:** U5.AOUT1 | **destination:** M_LEFT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

81. net: LEFT_FRONT_- | **origin:** U5.AOUT2 | **destination:** M_LEFT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

82. net: LEFT_FRONT_+ | **origin:** LEFT_FRONT_+ | **destination:** C_LEFT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

83. net: LEFT_FRONT_- | **origin:** LEFT_FRONT_- | **destination:** C_LEFT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

84. net: LEFT_REAR_+ | **origin:** U5.BOUT1 | **destination:** M_LEFT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

85. net: LEFT_REAR_- | **origin:** U5.BOUT2 | **destination:** M_LEFT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

86. net: LEFT_REAR_+ | **origin:** LEFT_REAR_+ | **destination:** C_LEFT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

87. net: LEFT_REAR_- | **origin:** LEFT_REAR_- | **destination:** C_LEFT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

88. net: RIGHT_FRONT_+ | **origin:** U6.AOUT1 | **destination:** M_RIGHT_FRONT.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

89. net: RIGHT_FRONT_- | **origin:** U6.AOUT2 | **destination:** M_RIGHT_FRONT.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

90. net: RIGHT_FRONT_+ | **origin:** RIGHT_FRONT_+ | **destination:** C_RIGHT_FRONT.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

91. net: RIGHT_FRONT_- | **origin:** RIGHT_FRONT_- | **destination:** C_RIGHT_FRONT.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

92. net: RIGHT_REAR_+ | **origin:** U6.BOUT1 | **destination:** M_RIGHT_REAR.+ | **wire awg:** 22 | **color:** yellow | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

93. net: RIGHT_REAR_- | **origin:** U6.BOUT2 | **destination:** M_RIGHT_REAR.- | **wire awg:** 22 | **color:** blue | **note:** Keep supplied 28AWG motor lead short; reverse both leads if forward test disagrees

94. net: RIGHT_REAR_+ | **origin:** RIGHT_REAR_+ | **destination:** C_RIGHT_REAR.1 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

95. net: RIGHT_REAR_- | **origin:** RIGHT_REAR_- | **destination:** C_RIGHT_REAR.2 | **wire awg:** component | **note:** 100nF ceramic >=25V across MOTOR tabs

96. net: PWM0 | **origin:** U10.PWM0 | **destination:** U8.2 | **wire awg:** 26 | **color:** signal

97. net: PWM0 | **origin:** PWM0 | **destination:** R20.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

98. net: GND | **origin:** GND | **destination:** R20.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

99. net: PWM5V_0 | **origin:** U8.3 | **destination:** R30.1 | **wire awg:** 26 | **color:** signal

100. net: SERVO_LEFT_YAW | **origin:** R30.2 | **destination:** SV0.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

101. net: SERVO_LEFT_YAW | **origin:** SERVO_LEFT_YAW | **destination:** R40.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

102. net: GND | **origin:** GND | **destination:** R40.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

103. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV0.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

104. net: GND | **origin:** J_SERVO.1 | **destination:** SV0.GND | **wire awg:** 22 | **color:** black/brown

105. net: PWM1 | **origin:** U10.PWM1 | **destination:** U8.5 | **wire awg:** 26 | **color:** signal

106. net: PWM1 | **origin:** PWM1 | **destination:** R21.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

107. net: GND | **origin:** GND | **destination:** R21.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

108. net: PWM5V_1 | **origin:** U8.6 | **destination:** R31.1 | **wire awg:** 26 | **color:** signal

109. net: SERVO_LEFT_PITCH | **origin:** R31.2 | **destination:** SV1.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

110. net: SERVO_LEFT_PITCH | **origin:** SERVO_LEFT_PITCH | **destination:** R41.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

111. net: GND | **origin:** GND | **destination:** R41.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

112. net: 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV1.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

113. net: GND | **origin:** J_SERVO.1 | **destination:** SV1.GND | **wire awg:** 22 | **color:** black/brown

114. net: PWM2 | **origin:** U10.PWM2 | **destination:** U8.9 | **wire awg:** 26 | **color:** signal

115. net: PWM2 | **origin:** PWM2 | **destination:** R22.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

116. net: GND | **origin:** GND | **destination:** R22.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

117. net: PWM5V_2 | **origin:** U8.8 | **destination:** R32.1 | **wire awg:** 26 | **color:** signal

118. **net:** SERVO_RIGHT_YAW | **origin:** R32.2 | **destination:** SV2.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

119. **net:** SERVO_RIGHT_YAW | **origin:** SERVO_RIGHT_YAW | **destination:** R42.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

120. **net:** GND | **origin:** GND | **destination:** R42.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

121. **net:** 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV2.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

122. **net:** GND | **origin:** J_SERVO.1 | **destination:** SV2.GND | **wire awg:** 22 | **color:** black/brown

123. **net:** PWM3 | **origin:** U10.PWM3 | **destination:** U8.12 | **wire awg:** 26 | **color:** signal

124. **net:** PWM3 | **origin:** PWM3 | **destination:** R23.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

125. **net:** GND | **origin:** GND | **destination:** R23.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

126. **net:** PWM5V_3 | **origin:** U8.11 | **destination:** R33.1 | **wire awg:** 26 | **color:** signal

127. **net:** SERVO_RIGHT_PITCH | **origin:** R33.2 | **destination:** SV3.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

128. **net:** SERVO_RIGHT_PITCH | **origin:** SERVO_RIGHT_PITCH | **destination:** R43.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

129. **net:** GND | **origin:** GND | **destination:** R43.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

130. **net:** 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV3.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

131. **net:** GND | **origin:** J_SERVO.1 | **destination:** SV3.GND | **wire awg:** 22 | **color:** black/brown

132. **net:** PWM4 | **origin:** U10.PWM4 | **destination:** U9.2 | **wire awg:** 26 | **color:** signal

133. **net:** PWM4 | **origin:** PWM4 | **destination:** R24.1 | **wire awg:** component | **note:** 10k pull-down at buffer input

134. **net:** GND | **origin:** GND | **destination:** R24.2 | **wire awg:** component | **note:** 10k pull-down at buffer input

135. **net:** PWM5V_4 | **origin:** U9.3 | **destination:** R34.1 | **wire awg:** 26 | **color:** signal

136. **net:** SERVO_HEAD | **origin:** R34.2 | **destination:** SV4.SIGNAL | **wire awg:** 26 | **color:** signal | **note:** 220ohm series resistor

137. **net:** SERVO_HEAD | **origin:** SERVO_HEAD | **destination:** R44.1 | **wire awg:** component | **note:** 10k pull-down at servo signal

138. **net:** GND | **origin:** GND | **destination:** R44.2 | **wire awg:** component | **note:** 10k pull-down at servo signal

139. **net:** 5V_SERVO | **origin:** J_SERVO.2 | **destination:** SV4.RED | **wire awg:** 22 | **color:** red | **note:** Individual branch; retain servo plug center positive

140. **net:** GND | **origin:** J_SERVO.1 | **destination:** SV4.GND | **wire awg:** 22 | **color:** black/brown

141. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.5 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

142. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.9 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

143. **net:** GND | **origin:** GROUND_STAR | **destination:** U9.12 | **wire awg:** 26 | **color:** black | **note:** Unused buffer INPUT tied low

144. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U9.4 | **wire awg:** 26 | **color:** red | **note:** Unused buffer OE high

145. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U9.10 | **wire awg:** 26 | **color:** red | **note:** Unused buffer OE high

146. **net:** 5V_SERVO | **origin:** U3.VOUT | **destination:** U9.13 | **wire awg:** 26 | **color:** red | **note:** Unused buffer OE high

147. **net:** MOTOR_IN | **origin:** MOTOR_IN | **destination:** C1.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

148. **net:** GND | **origin:** GND | **destination:** C1.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

149. **net:** SERVO_IN | **origin:** SERVO_IN | **destination:** C2.1 | **wire awg:** component | **note:** 220uF 25V low-ESR

150. **net:** GND | **origin:** GND | **destination:** C2.2 | **wire awg:** component | **note:** 220uF 25V low-ESR

151. **net:** LOGIC_IN | **origin:** LOGIC_IN | **destination:** C3.1 | **wire awg:** component | **note:** 100uF 25V

152. **net:** GND | **origin:** GND | **destination:** C3.2 | **wire awg:** component | **note:** 100uF 25V

153. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C4.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

154. **net:** GND | **origin:** GND | **destination:** C4.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

155. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** C5.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

156. **net:** GND | **origin:** GND | **destination:** C5.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

157. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** C6.1 | **wire awg:** component | **note:** 1000uF 10V low-ESR

158. **net:** GND | **origin:** GND | **destination:** C6.2 | **wire awg:** component | **note:** 1000uF 10V low-ESR

159. **net:** 5V_LOGIC | **origin:** 5V_LOGIC | **destination:** C7.1 | **wire awg:** component | **note:** 470uF 10V low-ESR

160. **net:** GND | **origin:** GND | **destination:** C7.2 | **wire awg:** component | **note:** 470uF 10V low-ESR

161. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** C8.1 | **wire awg:** component | **note:** 100nF ceramic

162. **net:** GND | **origin:** GND | **destination:** C8.2 | **wire awg:** component | **note:** 100nF ceramic

163. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** C9.1 | **wire awg:** component | **note:** 100nF ceramic

164. **net:** GND | **origin:** GND | **destination:** C9.2 | **wire awg:** component | **note:** 100nF ceramic

165. **net:** 5V_MOTOR | **origin:** 5V_MOTOR | **destination:** R15.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

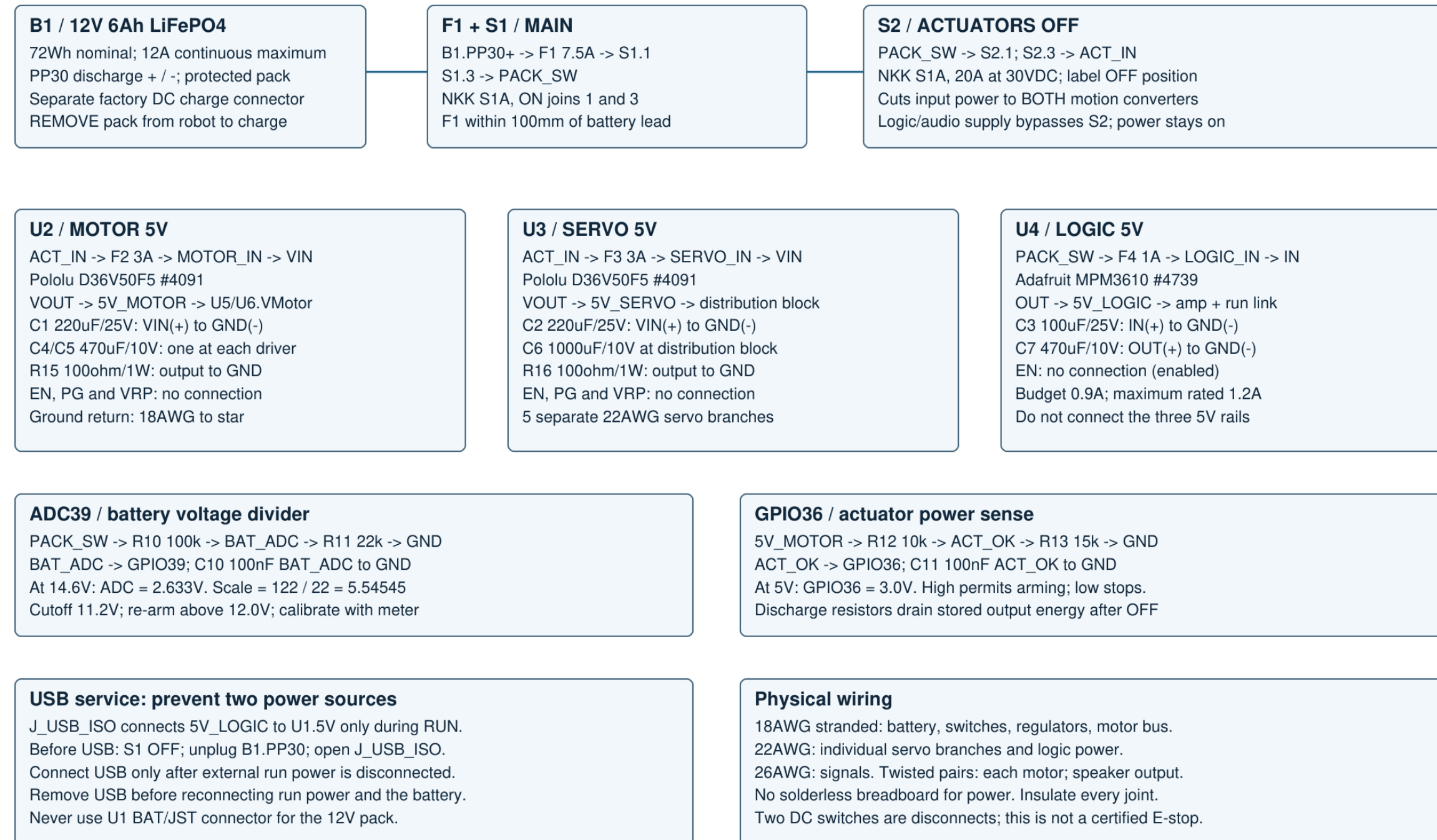
166. **net:** GND | **origin:** GND | **destination:** R15.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

167. **net:** 5V_SERVO | **origin:** 5V_SERVO | **destination:** R16.1 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

168. **net:** GND | **origin:** GND | **destination:** R16.2 | **wire awg:** component | **note:** 100ohm 1W discharge resistor

01 / Power distribution and isolation

B1 = Bioenno BLF-1206A + matched BPC-1502DC charger. All grounds join at GROUND_STAR.



02 / Controller, drive motors and sound

U1 = original LILYGO TTGO T-Display ESP32, 1.14 inch ST7789. GPIO numbers, not header positions.

U1 / exact external pin assignment

GPIO21	_____	SDA -> U10.SDA
GPIO22	_____	SCL -> U10.SCL
GPIO27	_____	SERVO_DISABLE / OE
GPIO12	_____	MOTOR_ENABLE / SLP
GPIO25	_____	LEFT_IN1
GPIO26	_____	LEFT_IN2
GPIO32	_____	RIGHT_IN1
GPIO33	_____	RIGHT_IN2
GPIO17	_____	I2S_BCLK
GPIO13	_____	I2S_LRC
GPIO15	_____	I2S_DATA
GPIO39 (SVN)	_____	BAT_ADC
GPIO36 (SVP)	_____	ACT_OK
3V3 / GND	_____	U10.VCC / common GND

Reserved board connections

TFT: MOSI19, SCLK18, CS5, DC16, RST23, BL4
Buttons: GPIO0 and GPIO35; reserve both.
Onboard battery ADC34 and power14 are NOT pack sense.
GPIO12 has 10k pull-down: required low boot strap state.
GPIO15 connects only to amplifier DIN; no external bias.
GPIO2, 37, 38 and UART pins: no external connection.

Hardware stopped on reset

R1..R4: 10k from each motor input net to GND.
R5: 10k MOTOR_ENABLE to GND; both SLP pins join it.
R6: 10k SERVO_DISABLE to3V3 (see sheet03).
Firmware: SLP low, inputs low, OE high before devices init.
Arm only after voltage, link and physical switch checks.
Physical power OFF works even if firmware stops responding.

U5 / Adafruit DRV8833 #3297

LEFT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
LEFT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_LEFT_FRONT +/-
BOUT1/BOUT2 -> M_LEFT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U6 / Adafruit DRV8833 #3297

RIGHT_IN1 -> AIN1 AND BIN1 (INPUTS ONLY)
RIGHT_IN2 -> AIN2 AND BIN2 (INPUTS ONLY)
SLP <- MOTOR_ENABLE; VMotor <- 5V_MOTOR; GND <- star
AOUT1/AOUT2 -> M_RIGHT_FRONT +/-
BOUT1/BOUT2 -> M_RIGHT_REAR +/-
Keep BOTH default0.2ohm sense resistors:1A per bridge.
DO NOT parallel the output bridges or two motors on one.
FLT, VM, ASEN, BSEN: no external connection.

U7 / MAX98357A + SPK1

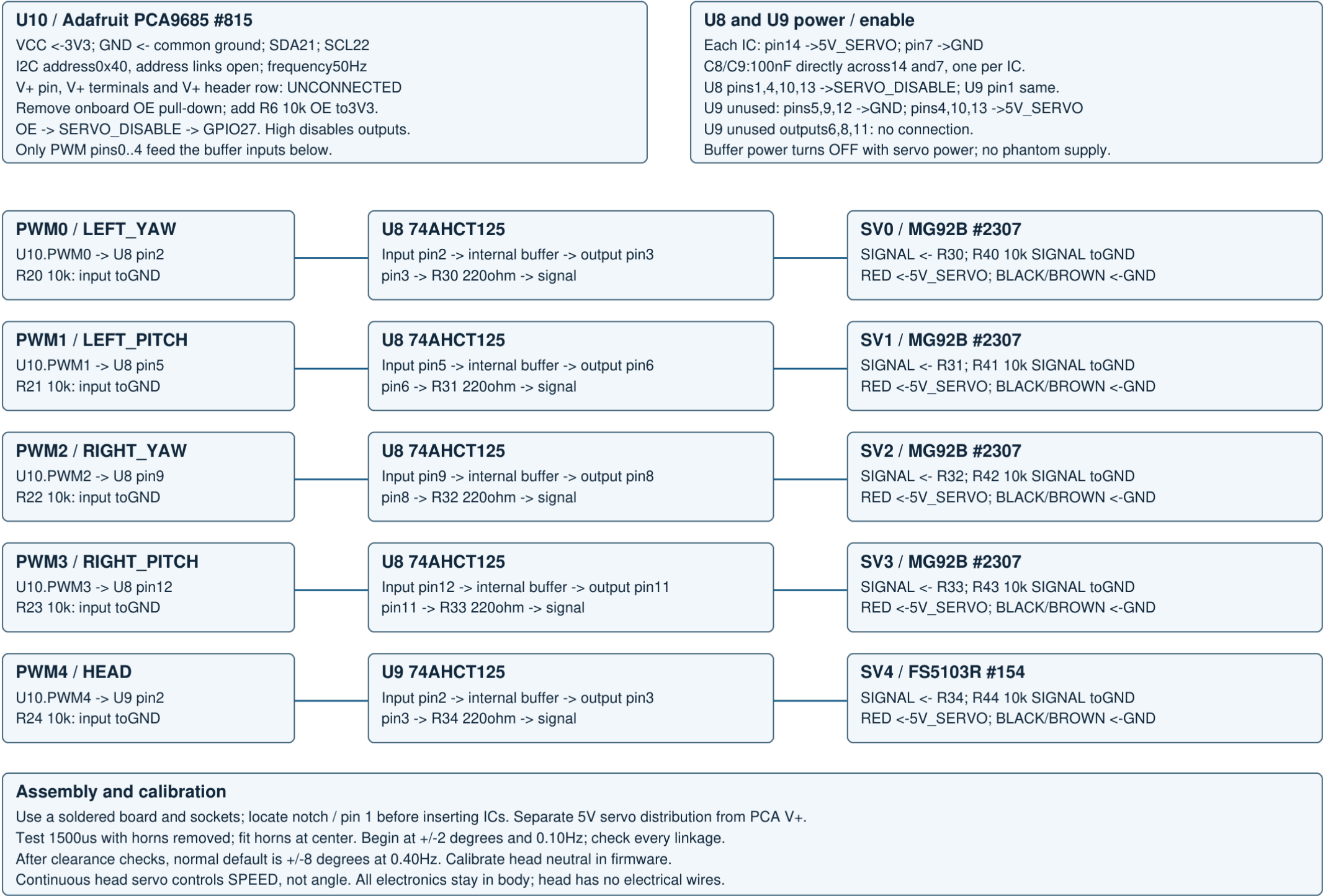
VIN <- 5V_LOGIC; GND <- common ground
BCLK <- GPIO17; LRC <- GPIO13; DIN <- GPIO15
GAIN -> R14 100k -> 5V_LOGIC (3dB)
SD: leave factory 1Mohm resistor configuration (mono mix)
OUT+ -> SPK1+; OUT- -> SPK1-; NEITHER lead to GND
SPK1: Adafruit 1313, 8ohm 1W; fixed digital gain 0.28
No external DAC, SD card or MP3 trigger module required.

Drive commissioning

100nF ceramic across each motor's metal tabs.
Use fast-decay PWM: one input PWM, other input low.
Stop both inputs low and SLP low; this is coast, not a brake.
Check each motor forward direction with wheels in air.
Swap that motor's two leads if its direction is wrong.
1A limit protects the driver, not indefinite motor stalls.
Indoor flat surface; no carpet, stairs or sustained wheel jams.

03 / Five servo channels with true 5V signals

All chip numbers below are DIP-14 PHYSICAL pin numbers. U8/U9 are 74AHCT125, not 74HC125.



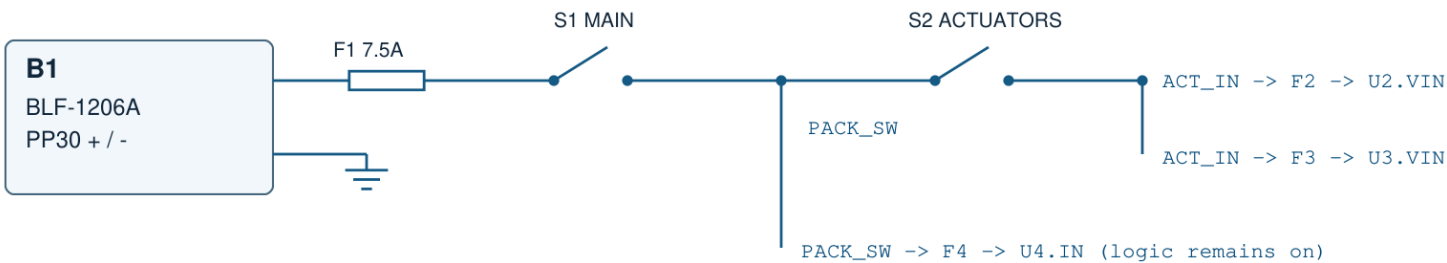
DALEK R1 | 2026-09-11 | Circuit + wiring.csv form one drawing set | Build validation is required

electronics\03-servos.svg | Drawing page: 17 x 11 inches. Use named nets and the wiring schedule together.

04 / Circuit details: power, sensing and bridge outputs

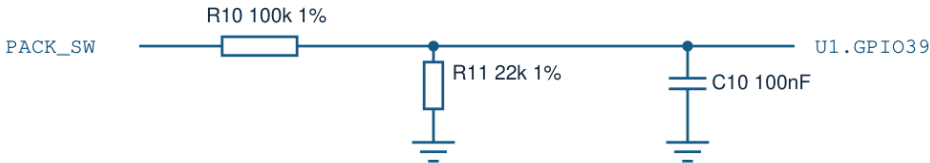
Conventional connection symbols. Dots are junctions. Matching net labels join all four sheets.

A. Battery fuse and independent motion disconnect



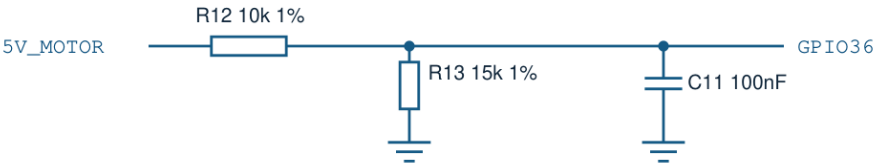
S1/S2 are shown OFF. NKK S1A ON joins terminals1 and3. Ground is never switched.

B. Battery measurement / GPIO39



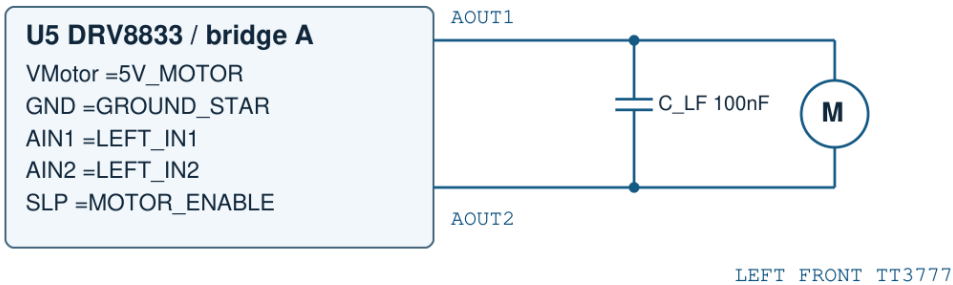
14.6V x22/(100+22) =2.633V; firmware multiplier5.54545

C. Motion power sense / GPIO36



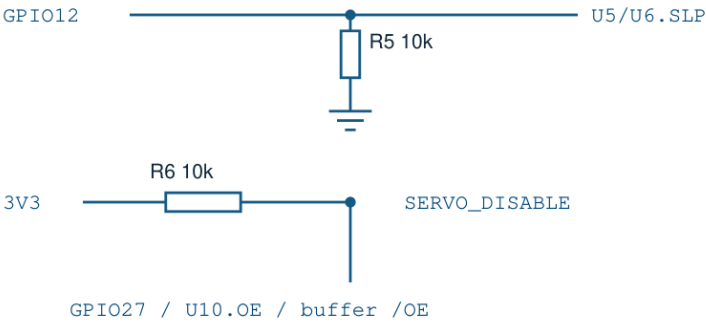
5V x15/(10+15) =3.0V; low means physical power OFF

D. Motor output circuit (repeat for all4 motors)



U5 bridgeB ->LEFT REAR; U6 bridgeA ->RIGHT FRONT; U6 bridgeB ->RIGHT REAR.
Bridge outputs never join one another. The only shared connections are INPUTS and supplies.

E. Reset default states



Remove originalPCAOE pull-down.
Repeat10k pull-down on each drive input.

POWER NOTES: F2/F3=3A; F4=1A. Correct capacitor polarity. All connection values also appear in sheets01-03.

DALEK R1 | 2026-09-11 | Circuit + wiring.csv form one drawing set | Build validation is required

electronics\04-circuit-details.svg | Drawing page: 17 x 11 inches. Use named nets and the wiring schedule together.